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Process Modelling & Simulation

Boundary $\rightarrow$ helps distinguish system & surroundings

system $\rightarrow$ where our interest is

closed system $\rightarrow$ not adiabatic

Isolated " $\rightarrow$ adiabatic

steady state vs Equilibrium

$\downarrow$

$\rightarrow$ backward & forward rates happen together

system does not change with time

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If rate of transport $\gg$ rate of accumulation

$\downarrow$

pseudo or quasi steady state

1D transient species transport eqn

$$\underbrace{\frac{\partial c}{\partial t}}_{\text{transient}} + \underbrace{u \frac{\partial c}{\partial x}}_{\text{convection}} = \underbrace{D \frac{\partial^2 c}{\partial x^2}}_{\text{diffusion}}$$

non dimensionalise the eqn

$$u^* = u/u_0 \quad c^* = c/c_0 \quad x^* = x/L \rightarrow \text{characteristic length scale}$$
$$L^2/D \rightarrow \text{char time scale}$$

$$\frac{\partial c^*}{\partial t^*} + \frac{u_0 u^*}{L} \frac{\partial c^*}{\partial x^*} = \frac{D}{L^2} \frac{\partial^2 c^*}{\partial x^{*2}}$$

$$t^* = \frac{tD}{L^2} = \frac{t}{t_c}$$

$$\frac{\partial c^*}{\partial t^*} + Pe \, u^* \frac{\partial c^*}{\partial x^*} = \frac{\partial^2 c^*}{\partial x^{*2}}$$

$$c^* \sim O(1)$$

$$u^* \sim O(1)$$

$$x^* \sim O(1)$$

if $t \gg \frac{L^2}{D} \rightarrow$ steady state

if $t^* \gg 1$ dynamics is not imp, system approaches (pseudo) ss behaviour

$$\frac{c^*}{t^*} \ll 1$$

$$\downarrow \quad t \gg \frac{L^2}{D} \rightarrow \frac{\partial c^*}{\partial t^*} \neq 0 \ll 1$$

above we multiplied the eqn by $\frac{L^2}{D}$, now we multiply with $\frac{L}{\mu_0}$

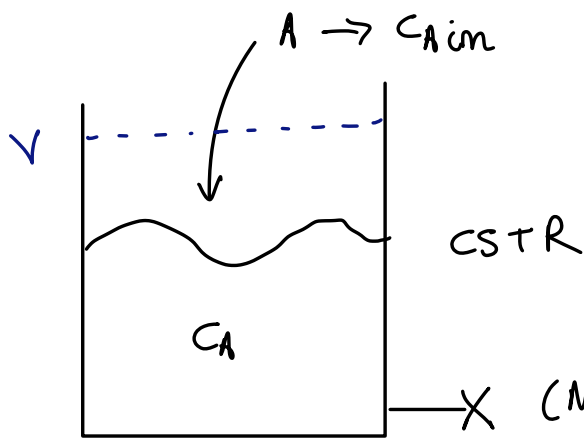
$$\frac{\partial c^*}{\partial t} + \frac{\mu_0 u^*}{L} \frac{\partial c^*}{\partial x^*} = \frac{D}{L^2} \frac{\partial^2 c^*}{\partial x^{*2}}$$

$$\frac{\partial c^*}{\partial t^*} + u^* \frac{\partial c^*}{\partial x^*} = \frac{1}{Pe} \frac{\partial^2 c^*}{\partial x^{*2}}$$

where $t^* = \frac{t \mu_0}{L}$ characteristic time scale $= \frac{L}{\mu_0}$

$Pe = \frac{\text{characteristic time scale of diff}}{\text{" " " " convection}}$

if physical time scale $\gg$ characteristic time scale $\rightarrow$ then we can drop the transient part and call it quasi steady state



In filling phase / startup phase,
there are some issues happening
How do you quantify it?



$0 \rightarrow t_1 \quad (= V_T / F)$

$V_T \rightarrow$ Volume of tank
 $F \rightarrow$ Inlet flow rate } so get time req. to fill the tank

$t_1 \rightarrow t_2$

$t_2 \rightarrow \infty$ (SS) $\rightarrow$ we write CSTR eqns from here
CSTR

$$C_A = \frac{C_{A0}}{1 + KC}$$

Mole balance (because V is changing)

$$\underbrace{\frac{d(C_A V)}{dt}}_{\text{accumulation}} = \underbrace{Q_{in} C_{Ain}}_{\text{Input}} - \underbrace{K C_A V}_{\text{Depletion}}$$

$$\frac{dm_A}{dt} = Q_{in} C_{Ain} - K m_A$$

$$\int_0^{m_A} \frac{dm_A}{Q_{in} C_{Ain} - K m_A} = \int_0^{t_1} dt$$

$$\left[\ln(Q_{in} C_{Ain} - K m_A) \right]_0^{m_A} = -K t$$

$$\frac{\ln(Q_{in} C_{Ain} - K m_A)}{Q_{in} C_{Ain}} = -K t$$

$$\frac{Q_{in} C_{Ain} - K m_A}{Q_{in} C_{Ain}} = \exp(-K t)$$

$$C_A V = m_A = \frac{Q_{in} C_{Ain}}{k} [1 - \exp(-kt)]$$

$$@ t \geq t_1, V = V_T$$

$$V_T \frac{\partial C_A}{\partial t} = Q_{in} C_{Ain} - Q_{out} C_A - k C_A V_T$$

will be non zero

[after reaching V_T , use open the outlet valve, to maintain the level]

$$\frac{\partial C_A}{\partial t} = \frac{Q_{in} C_{Ain}}{V_T} - \frac{Q_{out} C_A}{V_T} - k C_A$$

$$x_A = \frac{\partial C_A}{\partial t}$$

Reaction is assumed to be spontaneous (very fast, compared to transport of the species)
so process dynamics more imp than reaction

$$t_1 = V_T / Q$$

$$C_{A t_1} = C_A (@ t = t_1)$$

$$= \frac{Q_{in} C_{Ain}}{k V_T} [1 - \exp(-k \frac{V_T}{Q})]$$

$$C_A = \frac{C_{Ain}}{1 + k t_1} - \left[\frac{C_{Ain}}{1 + k t_1} - C_{A t_1} \right] \exp(-t(k + \frac{1}{t_1}))$$

$$C_{A t_1} = m_{A t_1} \times V_T$$

$$t_1 = \frac{V_T}{Q}$$

$$\frac{1}{k + \frac{1}{t_1}} \rightarrow \text{characteristic time scale}$$

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Classification of PDE's (1st or 2nd order)

$$\sum_{i,j} a_{ij} \underbrace{\frac{\partial u_i}{\partial x_i} \frac{\partial u_j}{\partial x_j}}_{\text{second order terms}} = f(u_i, x_i, \frac{\partial u_i}{\partial x_i})$$

$$\nabla^2 u$$

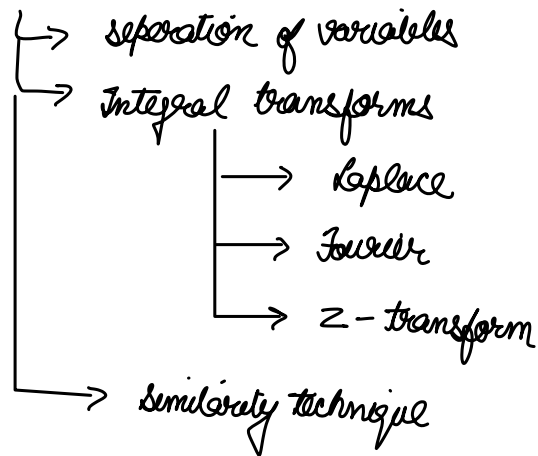
$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots \\ a_{21} & a_{22} & a_{23} & \dots \\ \vdots & & & \\ \vdots & & & \end{bmatrix}$$

Eigenvalue of this matrix a_{ij} $u_t + u_{xx} = 0$ at least one eigenvalue is zero $\rightarrow$ parabolic $u_{xx} + u_{yy} = 0$ all — of same sign $\rightarrow$ elliptic (time dependent patterns)
(multidimensional SS problem) $u_{xx} = u_{yy}$ different sign $\rightarrow$ hyperbolic (wave eqn, quantum mechanics)Parabolic $\rightarrow u_t + u_{xx} = 0$ Elliptic $\rightarrow u_{xx} + u_{yy}$ Hyperbolic $\rightarrow u_{tt} = u_{yy}$ $x + y + u_y = 0$ linear
$$\begin{aligned} x + u^2 + u_y \\ u_y + \sin u = 0 \end{aligned} \quad \left. \vphantom{\begin{aligned} x + u^2 + u_y \\ u_y + \sin u = 0 \end{aligned}} \right\} \text{Non linear}$$

Linear / Non linear

Non linear $\rightarrow$ analytical solutions are not possible (Numerical)

Analytical solution of POEs



Numerical solutions

Finite difference

" Volume

" element

Lattice Boltzmann method

Spectral decomposition

Orthogonal collocation

Separation of Variables

Parabolic PDE

1D unsteady heat eqn

$$u_t = \alpha u_{xx}$$

$$\hookrightarrow \frac{k}{\rho C_p}$$

BC

$$\textcircled{a} \quad x=0 \quad u=1$$
$$x=1 \quad u=0$$

IC

$$\textcircled{a} \quad t=0 \quad u=1$$

$$u_t = \alpha u_{xx}$$

consider $u = x(x) T(t)$

$$x T(t) = \alpha T x_{xx}$$

$$\underbrace{\frac{T_t}{T}}_{f(t)} = \alpha \underbrace{\frac{x_{xx}}{x}}_{f(x)} = \text{const} = -\lambda^2$$

$$\frac{T_t}{T} = -\lambda^2$$

$$x_{xx} = -\frac{\lambda^2 x}{\alpha}$$

$$T_t = -\lambda^2 T$$

$$T = c_3 \exp(-\lambda^2 t)$$

$$x = c_1 \sin\left(\frac{\lambda x}{\sqrt{\alpha}}\right) + c_2 \cos\left(\frac{\lambda x}{\sqrt{\alpha}}\right)$$

to prevent blowout of this term as $t \rightarrow \infty$
 $-\lambda^2$

Four constants $\rightarrow c_1, c_2, c_3, \lambda \rightarrow$ 4 eqns required
2 BC, 1 IC, 1 orthogonality

$$1 = c_2 \quad (x=0)$$

$$0 = c_1 \sin\left(\frac{1}{\sqrt{\lambda}}\right) + c_2 \cos\left(\frac{1}{\sqrt{\lambda}}\right)$$

Principle of orthogonality of trigonometric functions

$$\int \sin(\lambda_m x) \sin(\lambda_n x) dx = 0 \quad \text{if } m \neq n$$

$$u = \sum \text{const exp}(-\lambda^2 t) \left[(\cdot) \sin(\cdot) + (\cdot) \cos(\cdot) \right]$$

summation over
all possible eigenvalues of λ

Elliptic PDE

$$u_{xx} + u_{yy} = 0$$

$$u = X(x) Y(y)$$

$$u_{xx} = -u_{yy}$$

$$\frac{X_{xx}}{X} = -\frac{Y_{yy}}{Y} = \lambda^2$$

BCs

$$\textcircled{a} \quad x=0 \quad \begin{cases} u=0 \\ u=0 \end{cases}$$

$$y=0 \rightarrow u=1$$

$$y=1 \rightarrow u=0$$

$$\frac{d^2 X}{dx^2} = \lambda^2 X$$

$$X \frac{1}{X} \frac{d^2 X}{dx^2} = \lambda^2 dx$$

$$Y = c_1 \sin(\lambda y) + c_2 \cos(\lambda y)$$

$$Y = c_1 \exp(\lambda^2 y) + c_2 \exp(-\lambda^2 y)$$

$$+\lambda^2 \rightarrow Y_{yy} = -\lambda^2 Y$$

$$-\lambda^2 \rightarrow Y_{yy} = \lambda^2 Y$$

If we go with + sign $\rightarrow$ we will get trivial soln for X
so -ve sign

$$X = c_3 \exp(+i^2 x) + c_4 \exp(-i^2 x)$$

$$Y = c_1 \sin(i x) + c_2 \cos(i x)$$

can have same value
at different values

whichever dimension has homogeneous BCs, we should have trigonometric soln / periodic soln

" " " different BCs " " exponential soln
↓

because they are monotonic
so can't have same value at different
locations / values

Discretization (Finite Difference)

$\phi \rightarrow$ scalar valued function

$$\frac{\partial \phi}{\partial x} = \frac{\phi_{i+1} - \phi_i}{\Delta x} \quad (\text{Forward difference})$$

$$\frac{\phi_i - \phi_{i-1}}{\Delta x} \quad (\text{Backward difference})$$

$$\frac{\partial \phi}{\partial t} = \frac{\phi^{n+1} - \phi^n}{\Delta t}$$

Taylor series expansion

$$\phi_{i+1} = \phi_i + \frac{\partial \phi}{\partial x} \Delta x + \frac{\partial^2 \phi}{\partial x^2} \frac{(\Delta x)^2}{2!} + \frac{\partial^3 \phi}{\partial x^3} \frac{(\Delta x)^3}{3!} - \dots$$

Ignoring them

if $\Delta x \uparrow$, error will be higher

↓

We will compromise on accuracy of solution

answer is not that $\rightarrow$ Resolution will increase. Refined grids $\rightarrow$ more no.

but that is not the reason
behind use of refined grids

← of points

We do refinement so that Δx is small, error associated with ignored Taylor series terms is smaller

Second order terms (Central difference)

$$\frac{\partial^2 \phi}{\partial x^2} = \frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{(\Delta x)^2} \rightarrow \text{Need information from both points (forward \& backward)}$$

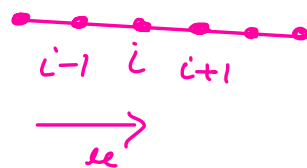
To get this substitute backward difference in Taylor series

providing convection $\leftarrow$ Courant Number Criteria (CFL) $\rightarrow$ 1st order PDE
von-Neumann Stability criteria (2nd order PDE)

$$\frac{\partial \phi}{\partial t} + \mu \frac{\partial \phi}{\partial x} = 0 \quad (\text{first order PDE})$$

$$\frac{\phi^{n+1} - \phi^n}{\Delta t} + \mu \frac{\phi_{i+1} - \phi_i}{\Delta x} = 0 \quad \phi_i - \phi_{i-1} \propto \Delta x$$

upwind scheme $\mu > 0$, use forward diff for $\partial \phi / \partial x$
 $\mu < 0$, " backward diff for $\partial \phi / \partial x$



$$\frac{\partial \phi}{\partial t} + \frac{\partial^2 \phi}{\partial t^2} \frac{\Delta t}{2} + \mu \frac{\partial \phi}{\partial x} + \mu \frac{\partial^2 \phi}{\partial x^2} \frac{\Delta x}{2} = 0$$

$$\frac{\partial \phi}{\partial t} + \mu \frac{\partial \phi}{\partial x} = -\frac{1}{2} \left(\Delta t \frac{\partial^2 \phi}{\partial t^2} + \mu \Delta x \frac{\partial^2 \phi}{\partial x^2} \right)$$

$$\frac{\partial \phi}{\partial t} = -\mu \frac{\partial \phi}{\partial x} \Rightarrow \frac{\partial^2 \phi}{\partial t^2} = -\mu \frac{\partial^2 \phi}{\partial x \partial t}$$

$$= \frac{1}{2} \left(1 - \frac{u \Delta t}{\Delta x} \right) \frac{\partial^2 \phi}{\partial x^2} u \Delta x \rightarrow \text{artificial diffusion}$$

↓

should always be +ve,
to stabilise the system

$$1 - \frac{u \Delta t}{\Delta x} \geq 0$$

$$\frac{u \Delta t}{\Delta x} \leq 1$$

↳ constant number CFL

Imp $\Rightarrow$ convection transported in one time step should not be too large that it crosses the grid.
size has to be small

$u \times$ time step shouldn't be large than grid size

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1st order PDE $\rightarrow$ artificial / numerical diffusion
(Courant number criterion)

$$\frac{|u| \Delta t}{\Delta x} \leq 1$$

2nd order PDE $\rightarrow$ parabolic PDE

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$$

1D transient fourier eqn

$N \rightarrow$ numerical solution (approximate)

$D \rightarrow$ exact solution

$$\varepsilon \equiv |N - D|$$

$\hookrightarrow$ error

$N \rightarrow$ satisfies

$D \rightarrow$ "

so ε should also satisfy

$$\frac{\mu_i^{N+1} - \mu_i^N}{\Delta t} = \alpha \frac{\mu_{i+1}^N - 2\mu_i^N + \mu_{i-1}^N}{(\Delta x)^2}$$

$N \rightarrow$ time (superscript)
 $\rightarrow$ space (subscript)

$$\frac{\varepsilon_i^{N+1} - \varepsilon_i^N}{\Delta t} = \alpha \frac{\varepsilon_{i+1}^N - 2\varepsilon_i^N + \varepsilon_{i-1}^N}{(\Delta x)^2}$$

$$\left| \frac{\varepsilon_i^{N+1}}{\varepsilon_i^N} \right| \leq 1$$

$\rightarrow$ for any space point, as we are marching in time the error should not grow in time, as we are advancing in time stamps (for convergence)

$\rightarrow$ as we progress in time, error should decrease

$$\varepsilon \rightarrow \sum_{km} e^{at} e^{ikm\pi x}$$

generally any finite (mesh) valued function can be decomposed into fourier series

$a \rightarrow$ amplitude

$km \rightarrow$ wave number

$\hookrightarrow$ function should be finite

fourier fns $\rightarrow$ periodic fns. sin, cos etc.

we can take out the summation $\rightarrow$ if 1 term satisfies, then others will also satisfy
 put the forward wave term in the equation

$$t \equiv N \Delta t$$

$$t + \Delta t \equiv (N+1) \Delta t$$

$$\frac{e^{i k_m (N+1) \Delta x} - e^{i k_m N \Delta x}}{\Delta t} = \alpha \left[e^{i k_m (N+1) \Delta x} - 2 e^{i k_m N \Delta x} + e^{i k_m (N-1) \Delta x} \right] / (\Delta x)^2$$

Divide $e^{i k_m N \Delta x}$

$$\frac{e^{i k_m \Delta x} - 1}{\Delta t} = \alpha \frac{e^{i k_m \Delta x} - 2 + e^{-i k_m \Delta x}}{(\Delta x)^2}$$

$$e^{i k_m \Delta x} = 1 + \frac{\alpha \Delta t}{(\Delta x)^2} [2 \cos(k_m \Delta x) - 2]$$

$$\left| \frac{\varepsilon^{N+1}}{\varepsilon^N} \right| = \left| 1 + \frac{2\alpha \Delta t}{(\Delta x)^2} [\cos(k_m \Delta x) - 1] \right|$$

$$= \left| 1 - \frac{4\alpha \Delta t}{(\Delta x)^2} \sin^2\left(\frac{k_m \Delta x}{2}\right) \right|$$

$$\left| 1 - \frac{4\alpha \Delta t}{(\Delta x)^2} \right| \leq 1$$

(0,2) range (0,1)

$$\frac{4\alpha \Delta t}{(\Delta x)^2} \leq 2$$

$$\boxed{\frac{\alpha \Delta t}{(\Delta x)^2} \leq \frac{1}{2}} \rightarrow \text{stability criterion for 2nd order PDE}$$

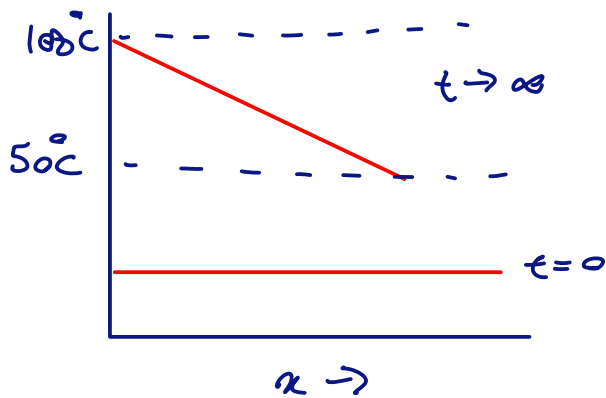
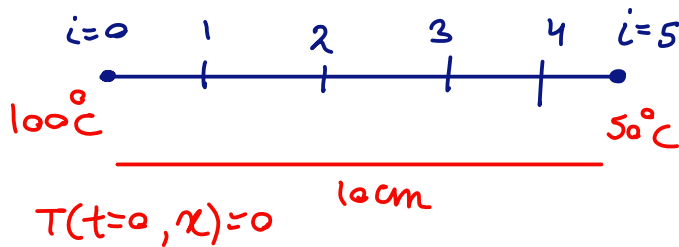
Finite difference $\rightarrow$ explicit scheme $\rightarrow$ stability scheme
 $\rightarrow$ implicit scheme $\rightarrow$ inherently stable
 $\rightarrow$ Crank Nicholson $\rightarrow$

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$$

$$\alpha = 0.835 \text{ cm}^2/\text{s}$$

$$L = 10 \text{ cm}$$

$$\Delta x = 2 \text{ cm}$$



1st let's check stability criterion

$$\frac{\alpha \Delta t}{(\Delta x)^2} = \frac{0.835 \times 0.1}{(2)^2}$$

$$= 0.020875 \leq \frac{1}{2}$$

$$\frac{u_i^{N+1} - u_i^N}{\Delta t} = \alpha \frac{u_{i+1}^N - 2u_i^N + u_{i-1}^N}{(\Delta x)^2}$$

$$u_i^{N+1} = u_i^N + \alpha \frac{(\Delta t)}{(\Delta x)^2} \{ u_{i+1}^N - 2u_i^N + u_{i-1}^N \}$$

$$\text{set } N=0 \quad i=(1,4)$$

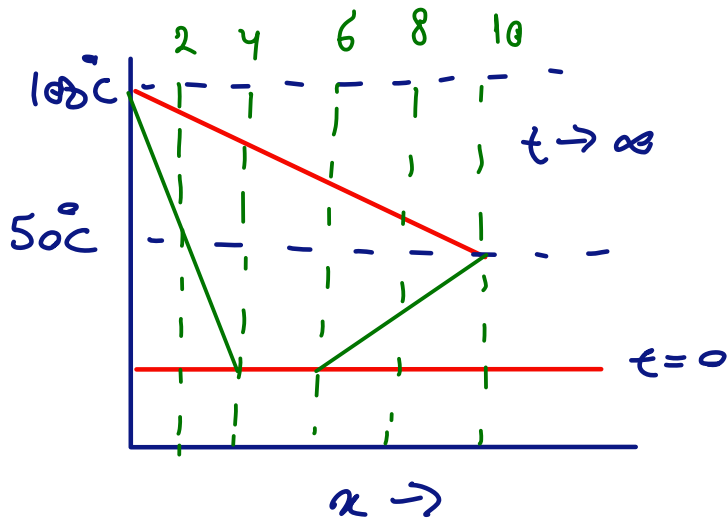
$$i=1 \quad u_1^1 = \cancel{u_1^0} + 0.020875 \{ \cancel{u_2^0} - 2\cancel{u_1^0} + \cancel{u_0^0} \} = 2.0875$$

$\underset{0}{\cancel{0}}$ $\underset{0}{\cancel{0}}$ $\underset{0}{\cancel{0}}$ $\underset{100}{\cancel{100}}$

$$i=2 \quad \mu_2' = \cancel{\mu_2^0} + 0.020875 \{ \cancel{\mu_3^0} - 2\cancel{\mu_2^0} + \cancel{\mu_1^0} \} = 0$$

$$i=3 \quad \mu_3' = 0$$

$$i=4 \quad \mu_4' = \cancel{\mu_4^0} + 0.020875 \{ \cancel{\mu_5^0} - 2\cancel{\mu_4^0} + \cancel{\mu_3^0} \} = 1.0438$$



(N=1)

$$\mu_i^2 = \mu_i^1 + 0.020875 \{ \mu_{i+1}^1 - 2\mu_i^1 + \mu_{i-1}^1 \}$$

$$\mu_1^2 = 4.0878$$

$$\mu_2^2 = 0.043577$$

$$\mu_3^2 = 0.021788$$

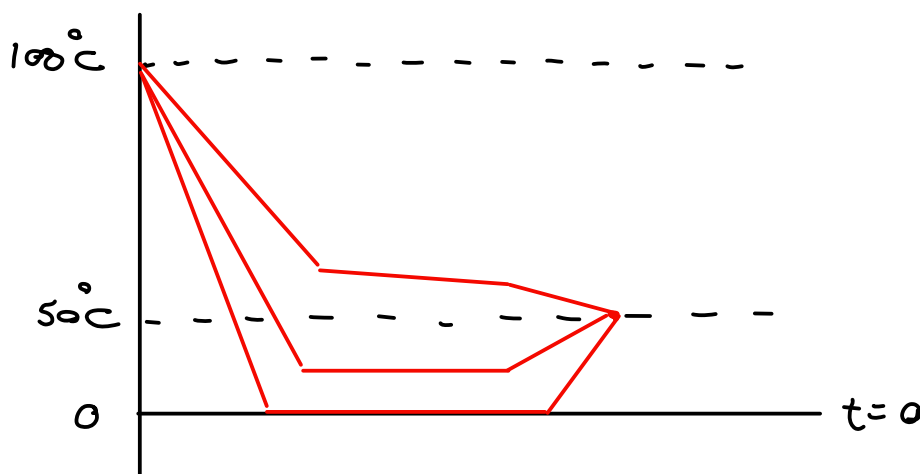
$$\mu_4^2 = 2.0439$$

① Boundary layer effect

② Dynamics of problem is imp

↳ how impact of BL is getting transmitted

③ Time scale at which solution approaches steady state solution



Implicit Scheme

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$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$$

FTCS

$$\frac{u_i^{N+1} - u_i^N}{\Delta t} = \alpha \frac{u_{i+1}^N - 2u_i^N + u_{i-1}^N}{(\Delta x)^2}$$

→ Explicit

Special discretization considered at previous time stamp

$$\frac{u_i^{N+1} - u_i^N}{\Delta t} = \alpha \frac{u_{i+1}^{N+1} - 2u_i^{N+1} + u_{i-1}^{N+1}}{(\Delta x)^2}$$

→ Special discretization considered at current (n+1) time step

⇓

Implicit

$$\underline{u_i^{N+1}} - \left(\alpha \frac{\Delta t}{(\Delta x)^2} \right) \left(\underline{u_{i+1}^{N+1}} - 2u_i^{N+1} + \underline{u_{i-1}^{N+1}} \right) = u_i^N$$

known

→ unknown

1 eqn, 3 unknown

$$\lambda u_{i+1}^{N+1} - (1+2\lambda) u_i^{N+1} + \lambda u_{i-1}^{N+1} = -u_i^N \quad i \in (1, m-1)$$

$$\begin{bmatrix} B_1 & C_1 & 0 & \dots & \dots & \dots \\ A_2 & B_2 & C_2 & 0 & \dots & \dots \\ 0 & A_3 & B_3 & C_3 & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \dots & \dots & \dots & A_{m-1} & B_{m-1} \end{bmatrix} \begin{bmatrix} u_1^{N+1} \\ u_2^{N+1} \\ \vdots \\ \vdots \\ u_{m-1}^{N+1} \end{bmatrix} = \begin{bmatrix} D_1 \\ D_2 \\ \vdots \\ \vdots \\ D_{m-1} \end{bmatrix}$$

$(m-1) \times (m-1)$

$$Ax = b$$

$$x = A^{-1}b$$

$$A_i = \lambda$$

$$B_i = -(1+2\lambda)$$

$$C_i = \lambda$$

$$D_i = -u_i^N$$

Tridiagonal matrix

$$\text{Size: } (m-1) \times (m-1)$$

$$\# \text{ non zero elements: } 3(m-1) - 2$$

$$\# \text{ total elements: } (m-1)^2$$

$$\text{Ratio} = \frac{3(m-1) - 2}{(m-1)^2}$$

$$\text{If } m \gg \text{large ratio} \sim \frac{1}{(m-1)}$$

→ every row has 3 elements except 1st & last row

Imp As you increase the size of the matrix, computation becomes slow and most of the calculations are dealing in zeroes, so time waste, so Thomas Algorithm

Thomas Algorithm

$$Ax = b$$

$$\begin{bmatrix} B_1 & C_1 & 0 & 0 & 0 \\ A_2 & B_2 & C_2 & 0 & 0 \\ 0 & A_3 & B_3 & C_3 & 0 \\ 0 & 0 & A_4 & B_4 & C_4 \\ 0 & 0 & 0 & A_5 & B_5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} D_1 \\ D_2 \\ D_3 \\ D_4 \\ D_5 \end{bmatrix}$$

$$A_i x_{i-1} + B_i x_i + C_i x_{i+1} = D_i$$

$$i=1$$

$$B_1 x_1 + C_1 x_2 = D_1$$

$$x_1 = \frac{D_1 - C_1 x_2}{B_1}$$

$$x_1 = p_1 - q_1 x_2$$

$$\text{where } p_1 = \frac{D_1}{B_1} \text{ \& } q_1 = \frac{C_1}{B_1}$$

$$x_1 + q_1 x_2 = p_1 \rightarrow 1 \leftarrow B_1, q_1 \leftarrow C_1 \text{ \& } p_1 \leftarrow D_1$$

$$\begin{bmatrix} 1 & q_1 & 0 & 0 & 0 \\ 0 & 1 & q_2 & 0 & 0 \\ 0 & 0 & 1 & q_3 & 0 \\ 0 & 0 & 0 & 1 & q_4 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \\ p_5 \end{bmatrix} \rightarrow \text{last row directly gives you } x_5$$

$i=2$

$$A_2 x_1 + B_2 x_2 + C_2 x_3 = D_2$$

$$A_2 (p_1 - q_1 x_2) + B_2 x_2 + C_2 x_3 = D_2$$

$$\Rightarrow x_2 + q_3 x_3 = p_2 \rightarrow \text{every equation includes current row \& next rows}$$

where

$$p_2 = \frac{D_2 - A_2 p_1}{B_2 - A_2 q_1}$$

$$q_2 = \frac{C_2}{B_2 - A_2 q_1}$$

$$l \leftarrow B_2$$

$$o \leftarrow A_2$$

$$q_2 \leftarrow C_2$$

$$p_2 \leftarrow D_2$$

last row ($i=5$)

$$x_5 = p_5$$

We start from the last row and backsubstitute

$$(i=4) \quad x_4 + q_4 x_5 = p_4$$

$\downarrow$
 p_5

$$x_3 + q_3 x_4 = p_4$$

Imp := No Inversion calculation, only focus on non-zero elements

Crank-Nicholson method

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$$

$$\frac{u_2^{N+1} - u^N}{\Delta t} = \frac{\alpha}{2} \left[\underbrace{\frac{u_{i+1}^N - 2u_i^N + u_{i-1}^N}{(\Delta x)^2}}_{\text{explicit}} + \underbrace{\frac{u_{i+1}^{N+1} - 2u_i^{N+1} + u_{i-1}^{N+1}}{(\Delta x)^2}}_{\text{Implicit}} \right]$$

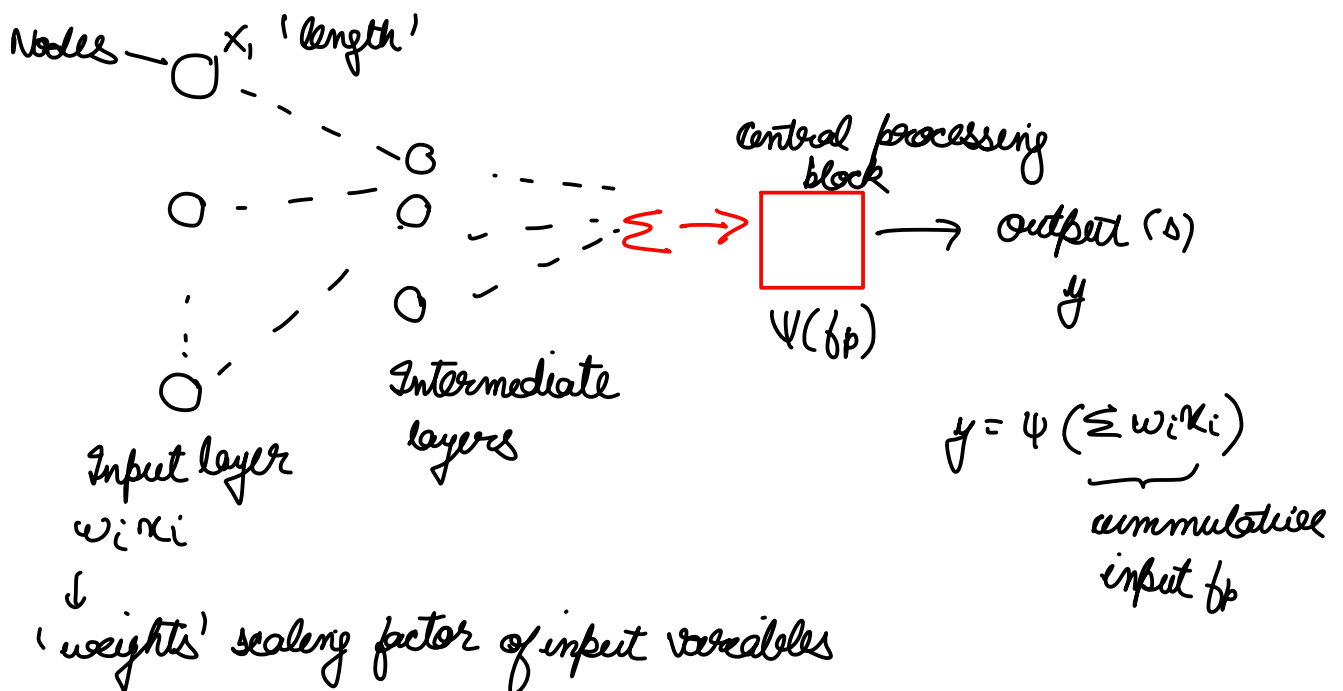
$$\Rightarrow -\lambda u_{i-1}^{N+1} + 2(1+\lambda) u_i^{N+1} - \lambda u_{i+1}^{N+1} = \lambda u_{i-1}^N + 2(1-\lambda) u_i^N + \lambda u_{i+1}^N$$

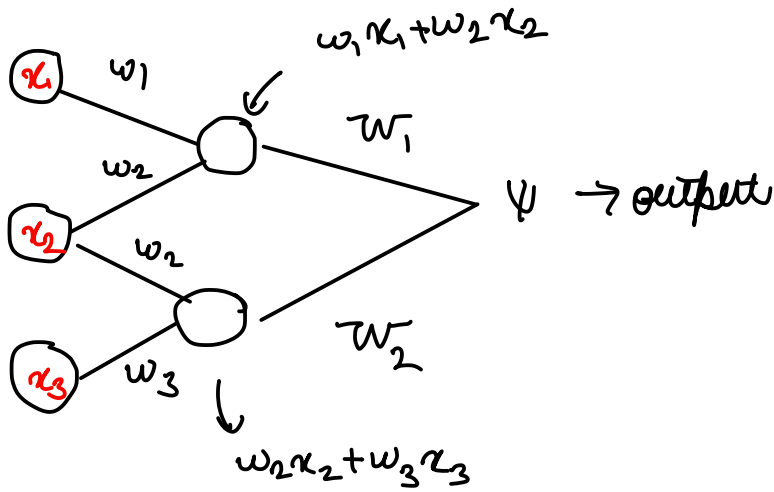
$$\text{where } \lambda = \alpha \frac{\Delta t}{(\Delta x)^2}$$

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Stochastic vs Deterministic Processes

Neural Networks





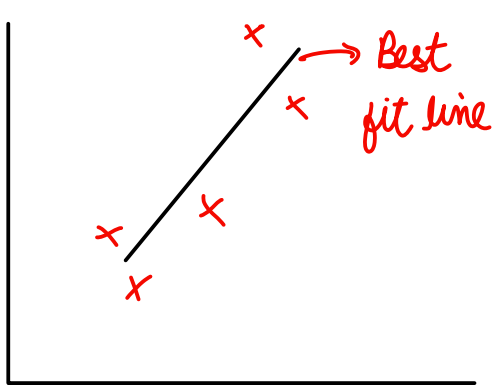
Types of ψ
 $\downarrow$
 linear, quadratic,
 gaussian, sigmoid etc.

$$\psi = \psi(W_1(w_1x_1 + w_2x_2) + W_2(w_2x_2 + w_3x_3))$$

Bias $\rightarrow$ Non zero input from zero input

$$y = \psi(\sum w_i x_i + \phi)$$

$\downarrow$
bias



$\rightarrow$ Linear regression / Multivariate regression

If more intermediate models present

$\downarrow$

Deep learning

[More evaluations of linear combinations needed]

estimation of weights $\rightarrow$ advanced regression [optimisation, minimisation of sum of squared errors b/w actual output and model output]

$$E = \frac{1}{N} \sum (t_p - b_p)^2$$

$\downarrow$ $\hookrightarrow$
actual output model output

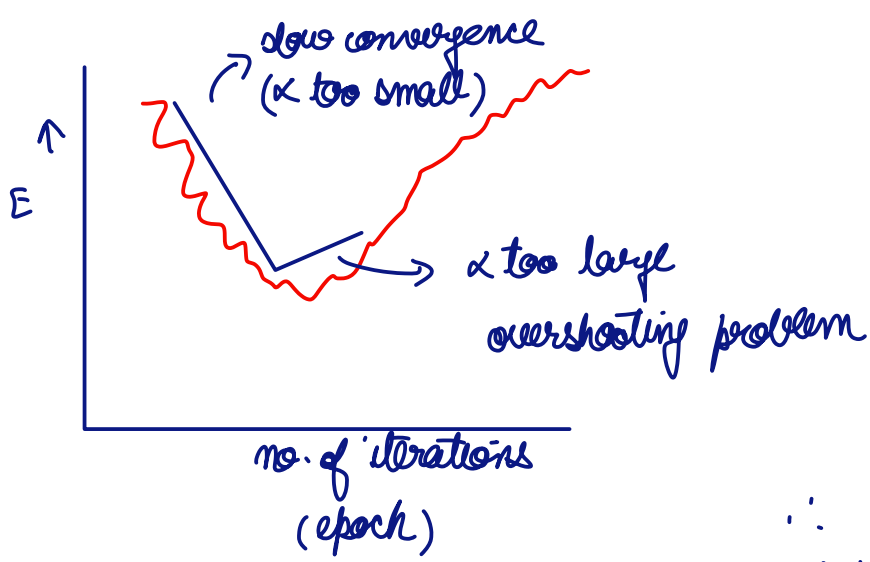
assuming output is scaled

$$\text{or } E = \frac{1}{N} \left(\frac{t_p - b_p}{t_p} \right)^2 \text{ if output is unscaled}$$

[Relative error matter]

Relative mean squared error

$$\psi = \psi(\sum w_i x_i + \phi) \quad w_i^{n+1} = w_i^n + \alpha \left(-\frac{\partial E}{\partial w_i} \right) \quad \alpha = \text{learning rate}$$



$$\frac{\partial E}{\partial w_i} = \frac{\partial E}{\partial y} \cdot \frac{\partial y}{\partial u} \cdot \frac{\partial u}{\partial w_i}$$

$$= -2(t_p - y) \underbrace{\frac{\partial y}{\partial u}}_{\psi'} x_i$$

$[u_i = w_i x_i + \phi]$

size of training data $\rightarrow$
at least 5-10 times no. of weights

$$N > \frac{|W|}{(1-a)}$$

$|W| \rightarrow$ no. of weights

$a \rightarrow$ expected accuracy on test set

Training, testing : (70:1 - 30:1. $\rightarrow$ good rule for training-testing on a given no. of data points)

$\therefore$

$$w_i^{N+1} = w_i^N + 2\alpha x_i (t_p - y) \psi'$$

$\psi' = 1 \rightarrow$ Widrow LMS method (linear)

$\psi' = y(1-y) \rightarrow$ generalised delta (sigmoid)

Distillation

→ Binary

→ Multi component → short cut method (FUG)

- Two variable approach
- Simultaneous correction
- Inside out
- Continuous methodology

components whose conc. is specified in distillate & bottom

key components

light key (LK)

↳ More volatile of

key components

heavy key (HK) → less volatile

of key components

N_{min} @ Total reflux

min no. of stages

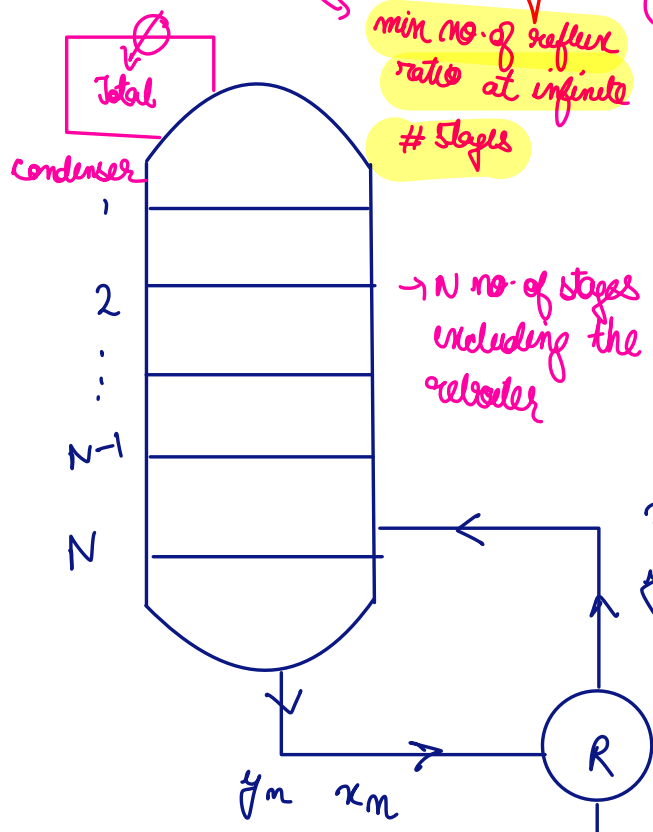
Fenske - Underwood - Gilliland (FUG)

min no. of reflux ratio at infinite

Stages

combo of both to get No. Reflux ratio

can be used for reboiler or n^{th} stage



→ N no. of stages including the reboiler

$$\left(\frac{Y_A}{Y_B} \right)_R = \alpha \left(\frac{x_A}{x_B} \right)_R \rightarrow \text{Relative volatility}$$

component mass balance around the reboiler

$$L = V + B$$

$$Lx_N = Vy_R + Bx_R$$

Component A → $y_{AR} = x_{AN}$ → everything entering in reboiler getting completely converted to vapour

Component B → $x_{BN} = y_{BR}$

Comp A → light key

Comp B → heavy key

liquid coming from previous tray is vaporising

$$x_{A,N-1} = y_{AN}$$

$$x_{B,N-1} = y_{BN}$$

$$\left(\frac{x_A}{x_B} \right)_{N-1} = \alpha_N \left(\frac{x_A}{x_B} \right)_N$$

$$\left. \frac{x_A}{x_B} \right|_N = \alpha_R \left. \frac{x_A}{x_B} \right|_R$$

$$\left. \frac{x_A}{x_B} \right|_{N-1} = \alpha_{N-1} \left. \frac{x_A}{x_B} \right|_R$$



You

total reflux in distillation



ChatGPT

Total reflux in distillation refers to a condition where all the vapor generated in the distillation process is condensed back into the distillation column and returned as liquid to the top of the column. In other words, no distillate product is withdrawn from the system during total reflux.

This condition is often employed in distillation operations for several purposes:

1. **Start-up:** Total reflux is commonly used during the start-up of a distillation column to establish steady-state conditions and to ensure proper temperature profiles throughout the column before product withdrawal begins.
2. **Optimization:** Total reflux can also be used for optimization purposes, allowing operators to adjust parameters such as reflux ratio, column temperatures, and tray efficiencies to achieve desired separation efficiency and product purity.
3. **Analysis:** Total reflux can be used for analytical purposes, such as determining the number of theoretical stages required for a given separation or assessing the performance of the distillation column under specific conditions.



In reboiler

$$\left(\frac{y_A}{y_B}\right)_R = \alpha \left(\frac{x_A}{x_B}\right)_R$$

$$\alpha_{AB} = \frac{y_A/x_A}{y_B/x_B}$$

→ continue till top tray

$$\left. \frac{x_A}{x_B} \right|_{\text{dist}} = \alpha_1 \alpha_2 \dots \alpha_{N-1} \alpha_N \alpha_R \left(\frac{x_A}{x_B} \right)_R$$

geometric average relative volatility
based on min # stages

$$\alpha_{AB} = (\alpha_1 \alpha_2 \dots \alpha_{N-1} \alpha_N \alpha_R)^{1/N_{\min}}$$

$$\left(\frac{x_A}{x_B} \right)_{\text{dist}} = \alpha_{AB}^{N_{\min}} \left(\frac{x_A}{x_B} \right)_R$$

$$N_{\min} = \frac{\ln(x_A/x_B)_{\text{dist}} - \ln(x_A/x_B)_R}{\ln(\alpha_{AB})}$$

In Fenske

we get min. no. of trays as we are assuming that we are operating in Total reflux

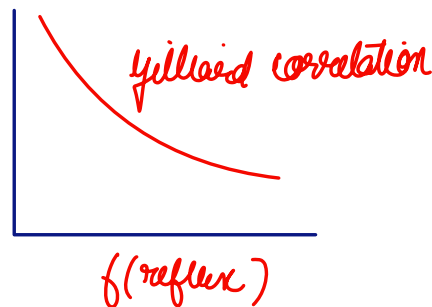
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Fenske → $N_{\min}$ (Total Reflux)

Underwood → Infinite # trays (consider min reflux)

Willard eqn → connects two eqns for some relating situations

f(# of trays)



Underwood → for min reflux condition

→ for which you require infinite # stages

at the feed location

$$x_{ij-1} = x_{ij} = x_{ij+1}$$

$$y_{ij+1} = K_i x_{ij+1}$$

$$y_{i,j-1} = y_{i,j} = y_{i,j+1}$$

Material balance in enriching section

section above the feed inlet

j → represents stage

i → represents component

$$V_{\min} y_{i,j+1} = L_{\min} x_{ij} + x_{i,\text{dist}} D$$

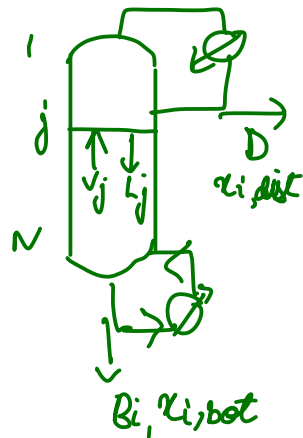
$$y_{i,j+1} = K x_{i,j+1} \rightarrow \text{from thermodynamics}$$

$$y_{i,j+1} = \frac{L_{\min} y_{ij}}{V_{\min} K} + \frac{x_d D}{V_{\min}}$$

$$y_{i,j+1} = \frac{L_{\min} y_{j+1}}{V_{\min} K} + \frac{x_d D}{V_{\min}}$$

$$\alpha = \frac{K}{K_{\text{ref}}} \quad K_i = \alpha_i K_{\text{ref}}$$

→ relative volatility



$$y_{i,j+1} = \frac{x_{i,D} D / V_{min}}{1 - \bar{L}_{min}} \cdot \frac{1}{V_{min} \alpha_i K_{ref}}$$

at intersection of top & bottom section of column

$$x_{ij-1} = x_{ij} = x_{ij+1}$$

↳ feed tray or composition of pinch tray

$$\sum_{i=1}^{\#} y_{i,j+1} = 1 = \sum_{i=1}^{\#} \frac{x_{i,D} D}{1 - \bar{L}_{min}} \cdot \frac{1}{V_{min} \alpha_i K_{ref}}$$

for stripping section → you can get same result as above

$$-1 = \sum \frac{x_{B,i} B / \bar{V}_{min}}{1 - \bar{L}_{min}} \cdot \frac{1}{\bar{V}_{min} \bar{\alpha}_i K_{ref}}$$

$$-\bar{V}_{min} = \sum \frac{\bar{\alpha}_i B x_{i,bot}}{\left[\bar{\alpha}_i - \bar{L}_{min} \right] \bar{V}_{min} K_{ref}}$$

enrich:

$$V_{min} = \sum \frac{x_{i,D} D}{1 - \bar{L}_{min}} \cdot \frac{1}{V_{min} \alpha_i K_{ref}} = \sum \frac{x_{i,D} D}{1 - \phi_i}$$

generally $\alpha_i = \bar{\alpha}_i$
 $K_{ref} \neq \bar{K}_{ref}$

stripping

$$-\bar{V}_{min} = \sum_{i=1}^C \frac{x_{i,B} B}{1 - \bar{L}_{min}} \cdot \frac{1}{\bar{V}_{min} \bar{\alpha}_i K_{ref}} = \sum \frac{x_{i,B} B}{1 - \phi_i}$$

$$V_{F,min} = \Delta V_{feed}$$

$$V_{min} - \bar{V}_{min} = \sum \frac{x_{i,D} D + x_{i,B} B}{1 - \phi_i}$$

1st Undenwood eqn

$$(1-q) F = \sum \frac{x_{i,D} D + x_{i,B} B}{1 - \phi_i}$$

↓
Quality

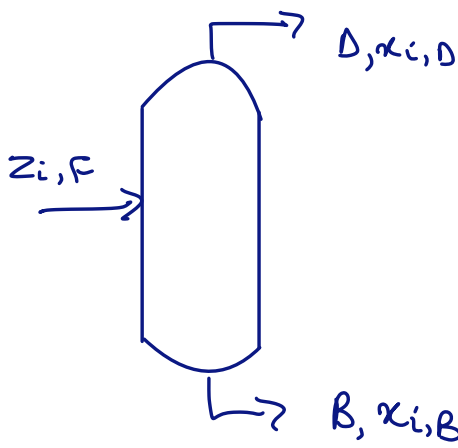
If we relax

$\alpha_{in} \leftarrow \alpha_i = \bar{\alpha}_i \rightarrow \alpha_{in}$ enriching section
stripping section $\phi = \bar{\phi}$

then we get

$$\Delta V_{feed} = 1-q$$

↳ ratio of amount of liquid component in feed



$$ZF = x_{i,D}D + x_{i,B}B$$

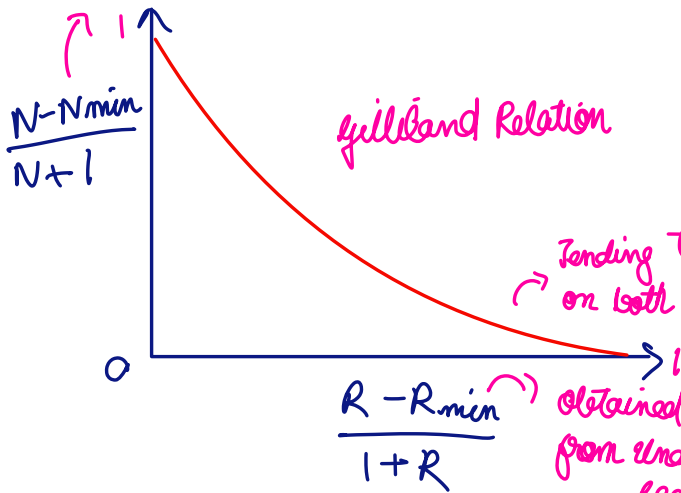
$$(1-q)F = \sum \frac{Z_i F}{1-\phi_i}$$

$$1-q = \sum \frac{Z_i}{1-\phi_i}$$

2nd Underwood eqn

rest known
can calculate
 ϕ
- (2)

obtained from Fenske eqn



Gilliland Relation

tending to ∞ ,
on both axes

$$\frac{R-R_{min}}{1+R}$$

obtained from Underwood eqn

get ϕ_i from (2)

$Q_i \rightarrow$ calculated - (2)

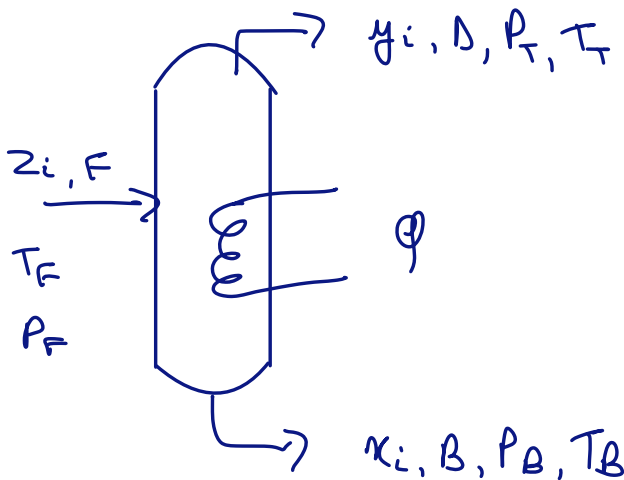
$V_{min} \rightarrow$ eqn (1) $\rightarrow$ get V_{min} from (1)

$$L_{min} = V_{min} - D$$

$$R_{min} = \frac{L_{min}}{D} \quad [\text{Target}]$$

$N \rightarrow$ # trays inside the column (not reboiler)

Flash Distillation



variable

$$3(C+3) + 1 \rightarrow 3C+10$$

three stream variable in each stream ϕ

$$\# \text{ equations} \rightarrow 2C+5$$

$$Z_i F = y_i V + x_i L$$

$$\text{comp material balance} \rightarrow C$$

$$\text{Thermal eq}^m \rightarrow 1 \quad [T_V = T_L]$$

$$\text{Press eq}^m \rightarrow 1 \quad [P_V = P_L]$$

$$\text{Phase eq}^m \rightarrow C \quad [y_i = K x_i]$$

Mech eqn

Thermal

Overall

comp. mass balance

Energy summation

MTP

over

complete

Overall material $\rightarrow 1$ $[F=V+L]$

balance

$$Q + h_F F = h_L L + h_V V$$

← energy balance $\rightarrow 1$

$$\# \text{ eqns} = 2C + 5$$

$$\text{summation} \rightarrow \sum x_i - \sum y_i = 0 \rightarrow 1$$

known parameter, feed parameter $\rightarrow C+3 \rightarrow [z_i, F, T_F, P_F]$

$$\text{unknown} \rightarrow 3C+10 - (C+3) = 2C+7$$

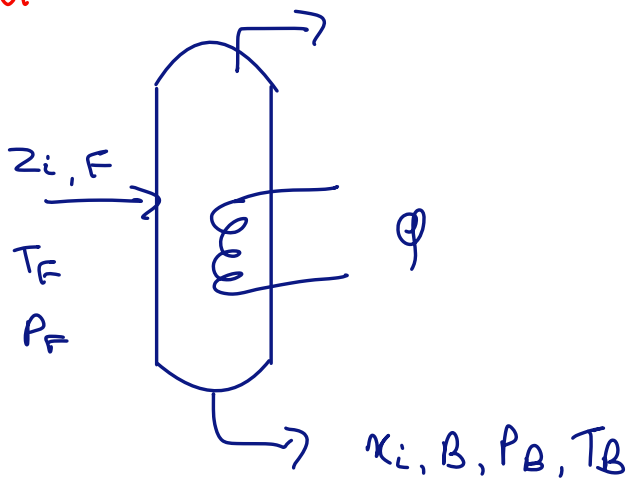
$\rightarrow C+3$ specified

$$\text{DOF} \rightarrow 2C+7 - (2C+5) = 2 \rightarrow 2 \text{ conditions has to}$$

$\rightarrow \# \text{ variable} \rightarrow \# \text{ eqn}$

be specified to solve

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$$3(C+3)+1 \rightarrow 3C+10$$

eqns $\rightarrow$ component mass balance

$$\hookrightarrow z_{i,F} F = y_{i,D} D + x_{i,B} B \quad (C)$$

$$\text{Thermal eqn} : T_D = T_B \quad (1)$$

$$\text{Press eqn} : P_D = P_B \quad (1)$$

$$\text{Phase Equil} : y_{i,D} = K x_{i,B} \quad (C)$$

$$\text{enthalpy} : \underline{h_F} F = h_D D + h_B B \quad (1')$$

$\downarrow$

If we know, $T \& P \rightarrow$ this value is known

Overall mass balance (1)

$$F = D + B$$

summation $\sum x_i = 1$ $\sum y_i = 1$

If these two are used then overall mass balance is reduced

$$\sum x_i - \sum y_i = 0 \quad (1)$$

$$\# \text{ eq} \rightarrow 2C + 5$$

$$\# \text{ Unknowns vars} : (3C + 10) - \underbrace{(C + 3)}_{\text{feed variables}} = 2C + 7$$

DOF : $(2C + 7) - (2C + 5) = 2$

Implicitly $(x_i = z_i)$ $D/F = 0$, T_D or $P_D \rightarrow$ Bubble point flash [Nothing from top] No vapour all liquid

Implicitly $(y_i = z_i)$ $D/F = 1$, T_B or $P_B \rightarrow$ Dew point flash [Nothing from B] Vapours everything in vapour phase

$T_D, P_D \rightarrow$ Isothermal flash $\rightarrow T_V, P_V$ given

$Q = 0$ T or $P \rightarrow$ adiabatic flash

$Q \neq 0$ T or $P \rightarrow$ non adiabatic flash

Isothermal case (T & P) Rachford rice method

$$z_i F = y_i D + x_i B \quad ; \quad F = D + B$$

$$z_i = y_i \frac{D}{F} + x_i \frac{B}{F} \quad 1 = \frac{D}{F} + \frac{B}{F}$$

$\hookrightarrow K_i x_i$

$$z_i = \left(K_i \frac{D}{F} + \frac{B}{F} \right) x_i$$

Bubble point

$\hookrightarrow$ first bubble forms

Dew point

$\hookrightarrow$ first dew/condensate forms

Dew point > Bubble point

$$x_i \left(\frac{D}{F} (K_i - 1) + 1 \right) = z_i$$

$$x_i = \frac{z_i}{1 - \frac{D}{F} (1 - K_i)}$$

$\frac{D}{F} \rightarrow \theta$

$$x_i = \frac{z_i}{1 - \theta (1 - K_i)}$$

$$y_i = \frac{K_i z_i}{1 - \theta (1 - K_i)} \rightarrow \text{as } y_i = K_i x_i$$

$$\frac{\theta_{m+1} - \theta_m}{\theta_m} < \epsilon$$

$f(\theta) < \epsilon_2$

both conditions to be fulfilled for convergence

Using summation $[\sum x_i - \sum y_i = 0]$

$$\sum \frac{(1 - K_i) z_i}{1 - \theta (1 - K_i)} = 0 \Rightarrow \sum_{i=1} \frac{(1 - K_i) z_i}{1 - \theta (1 - K_i)} = 0$$

$f(T, P)$ feed (known)

θ is unknown

Roots of summation equation above, gives you ' θ ' value

Rackford - Reice convergent method

Special case

Bubble point flash: $\theta = 0$ $D/F = 0$

$$\sum z_i (1 - K_i) = 0$$

$$\sum z_i - \sum z_i K_i = 0$$

1

$$\sum z_i K_i = 1 \rightarrow f(T, P)$$

Dew point flash $\theta = 1$ | $D/F = 1$

$$\sum \frac{z_i (1 - K_i)}{K_i}$$

$K \rightarrow f(T, P)$
 so, T, P of the system needs to be known

$$\sum \frac{z_i}{K_i} - \sum z_i = 0$$

$$\sum \frac{z_i}{K_i} - 1 = 0$$

$\rightarrow F(T, P) \rightarrow$

pressure will be specified, only temp will be unknown and so you can calculate dew pt temperature

Pressure $\rightarrow$ known

$$h_F F + Q = h_D D + h_B B$$

$$h_F + \frac{Q}{F} = h_D \theta + h_B (1 - \theta)$$

$\uparrow$
known
 $\searrow f(\theta)$

$$\sum_{i=1}^C \frac{z_i (1 - K_i)}{1 + \theta (K_i - 1)}$$

$\downarrow$
 $f(T, P)$

$\rightarrow$ known

Temp $\rightarrow$ unknown

$\rightarrow$ it does not come directly into the picture (it comes with using

$h(T, K_i)$

3. adiabatic flash ($Q = 0$)

or non adiabatic flash ($Q \neq 0$) $\left\{ \begin{array}{l} \text{when } Q \text{ is given} \\ [1 \text{ DOF}] \end{array} \right.$

Energy balance: $h_F F + Q = h_V V + h_L L$

$$h_F + \frac{Q}{F} = h_V \theta + h_L (1 - \theta)$$

only θ, T_V unknown
 $\rightarrow$ as h_V, h_L depend on T_V

$$\leftarrow f(T_V) = \theta h_V + (1 - \theta) h_L - h_F + \frac{Q}{F} = 0 \quad \text{on } T_V$$

generally $h_L, h_V = f(T_V)$

& $K_i = f(T_V, P_V)$

$\left\{ \begin{array}{l} \text{don't know} \\ \text{exact, but know} \\ K_i \text{ it is a fn.} \end{array} \right.$

$\rightarrow f(T_V, P_V)$

$\rightarrow$ specified, so here also K_i depends on T_V

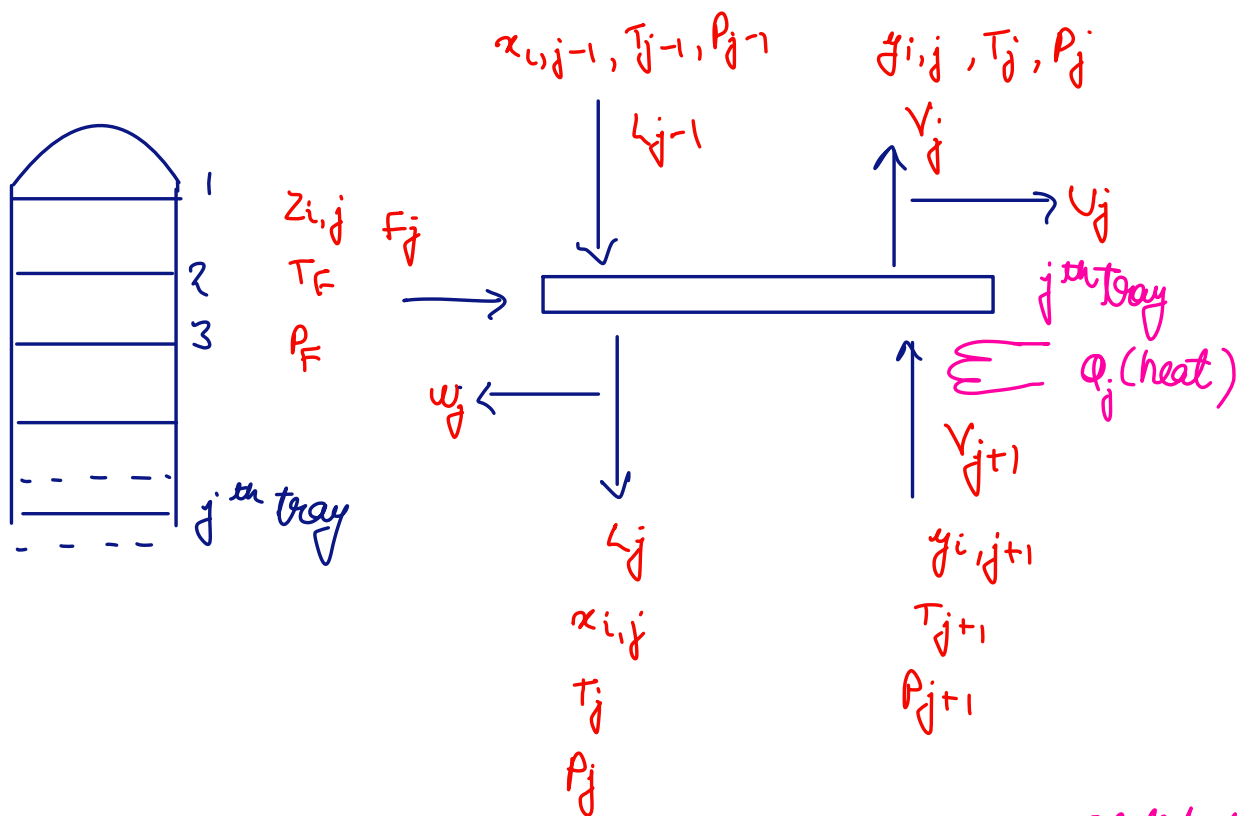
so 2 eqns with 2 unknowns effectively θ & T_V

for θ
we already
have

$$f(\theta) = \sum_{i=1}^C \frac{z_i (1 - K_i)}{1 + \theta (K_i - 1)} = 0$$

Multistage Multicomponent

drawing water $\rightarrow j^{th}$



Equation

Comp Material balance liquid in vapour in

$$M_{ij} \equiv \underbrace{Z_{i,j} F_j}_{\text{Feed in}} + \underbrace{x_{i,j-1} L_{j-1}}_{\text{liquid in}} + \underbrace{y_{i,j+1} V_{j+1}}_{\text{vapour in}}$$

$$- \underbrace{(L_j + w_j) x_{i,j}}_{\text{liquid out}} - \underbrace{(V_j + U_j) y_{i,j}}_{\text{vapour out}} = 0$$

Equilibrium relation

$$y_{i,j} = K_{i,j} x_{i,j} \rightarrow c \text{ equations per tray}$$

S- summation eqn 2 eqn per tray

$$(Sx)_j \equiv \sum_{i=1}^c x_{i,j} = 1$$

these used don't use overall material balance so can use two summation eqns!!!

$$(Sy)_j \equiv \sum_{i=1}^c y_{i,j} = 1$$

H-enthalpy eqn

$$h_F F + h_{L,j-1} L_{j-1} + h_{V,j+1} V_{j+1} - (V_j + U_j) h_{V,j} - (L_j + w_j) h_{L,j} + Q = 0$$

$\hookrightarrow 1 \text{ eqn per tray}$

MESH eqn $\rightarrow (2C+3)$ for each stage

$\downarrow \downarrow \downarrow \downarrow$
C C 2 1

$\hookrightarrow$ for n trays $n(2C+3)$ total eqns

M $\rightarrow$ Material balance

E $\rightarrow$ eq. relations

S $\rightarrow$ summation eqns

H $\rightarrow$ Energy balance

Variables $\rightarrow 5C+16$ / tray
 $3F, 6I, 6out, 1Q$

$\rightarrow$ not considered
 $h \rightarrow$ of enthalpies
 as they are related to T & P, so not explicit variables

Knowns $\rightarrow (C+3) \rightarrow$ feed

$2(C+3) \rightarrow$ for mixing trays $\rightarrow$ coming from previous streams
 $\parallel$
 $3C+9$

Unknown $\Rightarrow (5C+16) - (3C+9) \Rightarrow 2C+7$

DOF $\Rightarrow \underbrace{2C+7}_{\text{Unknown}} - \underbrace{(2C+3)}_{\text{eqn}} = 4$ # var to be calculated
 4 DOF per stage

$4 \rightarrow u_j, w_j, Q_j, \begin{matrix} x_{i,j} \\ y_{i,j} \\ T_j \\ P_j \\ L_j \\ V_j \end{matrix}$
 specified generally,
 so only one
 more quantity required
 to be known
 so that DOF becomes 0

one of them has to be specified
 (V_j, L_j, T_j, P_j) for solving. So that's where we take a guess on one of the values & iterations
 $\rightarrow$ If we use this $\rightarrow$ Bubble point or Dew point
 operating as
 Not Double
 Come into picture

ME eqn

$\hookrightarrow ME_{i,j}$

$$Z_{i,j} F_j + x_{i,j-1} L_{j-1} + K_{i,j+1} x_{i,j+1} V_{j+1} - (V_j + U_j) K_{i,j} x_{i,j} - (L_j + W_j) x_{i,j} = 0$$

$$L_{j-1} x_{i,j-1} - [(V_j + U_j) K_{i,j} + (L_j + W_j)] x_{i,j} + K_{i,j+1} V_{j+1} x_{i,j+1} = -Z_{i,j} F_j$$

$$ME_{i,1} \rightarrow L_0 x_{i,0} - [(V_j + v_j) K_{0,j} + (L_1 + W_1)] x_{i,1} + K_{0,2} V_2 x_{i,2} = 0$$

Total condenser

$$(ME)_{ij} = (1 + \gamma_{vj}) V_j K_{ij} x_{ij} + (1 + \gamma_{lj}) L_j x_{ij} - V_{j+1} K_{ij+1} x_{ij+1} - L_{j-1} x_{i,j-1} - F_j z_{ij} = 0$$

$$ME_{i,1} \equiv -(L + D) x_{i,1} + K_{i,2} V_2 x_{i,2} = 0$$

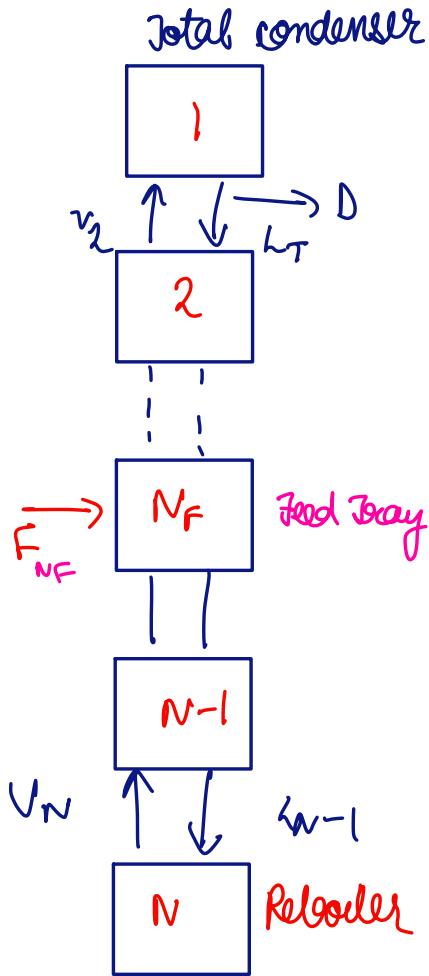
$\gamma_{vj} = 0, \gamma_{lj} = 0, F_j = 0$ [for 1st stage]

$$ME_{i,NF} \equiv LHS = -Z_{i,NF}, F_{NF}$$

$j = NF$

$$ME_{i,N} \equiv +L_{N-1} x_{i,N-1} - [V_N K_{i,N} + B] x_{i,N} = 0$$

↓
For reboiler



Physical # trays inside the columns

(For 2 to $N_F - 1$) & ($N_F + 1$ to $(N - 1)$)

$\forall j$ except $N_F, F_j = 0$

$\forall j$ " 1, N, V_j & $W_j = 0$

$\gamma_{vj} = 0$
 $\gamma_{lj} = F_j = 0$

$\gamma_{vj} = \frac{V_j}{V_j}$

$\gamma_{lj} = W_j / L_j$

→ (B) (upward)

generalising $(ME)_{ij}$ as $(ME)_{ij} = A_{ij} x_{ij-1} + B_{ij} x_{ij} + C_{ij} x_{ij+1} = D_{ij}$

$$\begin{bmatrix} B_{i,1} & C_{i,1} & 0 & 0 & \dots & 0 \\ A_{i,2} & B_{i,2} & C_{i,2} & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \dots & \dots & \dots & A_{i,N} & B_{i,N} \end{bmatrix} \begin{bmatrix} x_{i,1} \\ x_{i,2} \\ \vdots \\ \vdots \\ x_{i,N} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ -D_{i,N_F} \\ \vdots \\ 0 \end{bmatrix}_N$$

$A_{i,N-1} \ B_{i,N-1} \ C_{i,N-1}$

$N \times N$

$$A_{ij} = L_j - 1 \quad j = 2 \text{ to } N$$

$$B_{i,j} = -[(V_j + U_j) K_{i,j} + (L_j + W_j)] \quad j = 1 \text{ to } N$$

$$A_{ij} = L_j - 1$$

$$B_{ij} = -[1 + V_j] K_{ij} V_j + (1 + L_j) L_j$$

$$C_{ij} = K_{i,j+1} V_{j+1}$$

$$D_{ij} = -Z_{i,j} F_j$$

$$C_{ij} = V_{j+1} K_{i,j+1}$$

$$D_{i,j} = -Z_{i,j} F_j$$

Solution algorithm (bubble point calculation)

steps

① Assume values of V_j & T_j

Tray variable (?)

& Fix u_j & w_j

unknowns (fix)

$T_j, P_j, V_j, L_j, x_{ij},$

y_{ij}, u_j & w_j

② Calculate L_j from material balance

③ Calculate / estimate $K_{i,j}$ (T_j & P_j)

④ Solve TDM for each species $x_{ij} \rightarrow \sum_{i=1}^C x_{ij} \neq 1$

⑤ Normalise $\bar{x}_{ij} = \frac{x_{ij}}{\sum_{i=1}^C x_{ij}}$

↪ solve using Thomas algorithm

Ensures $\rightarrow \sum_{i=1}^C \bar{x}_{ij} = 1$

$\sum x_{ij} = 1 \rightarrow$ should be but our solution of Thomas algorithm is based on guess of T_j, V_j so $\sum x_{ij} \neq 1$ might be possible, so we normalise, to ensure $\sum \bar{x}_{ij} = 1$

⑥ Calculate the bubble pt. temp. for each tray

$$\sum_{i=1}^C y_{i,j} = 1 \Rightarrow \sum_{i=1}^C K_{i,j} \bar{x}_{ij} = 1 \quad \delta(T_j)$$

↪ use normalised tray composition

$y = Kx \rightarrow$ phase equilibrium

⑦ Since T_j & $\bar{x}_{ij}$ are known, calculate V_j (Vapour flow rate) in each from 'H' equation
Use enthalpy eqn, H eqn to back calculate $V_j \rightarrow$ compute h_j

from T_j
of step 5

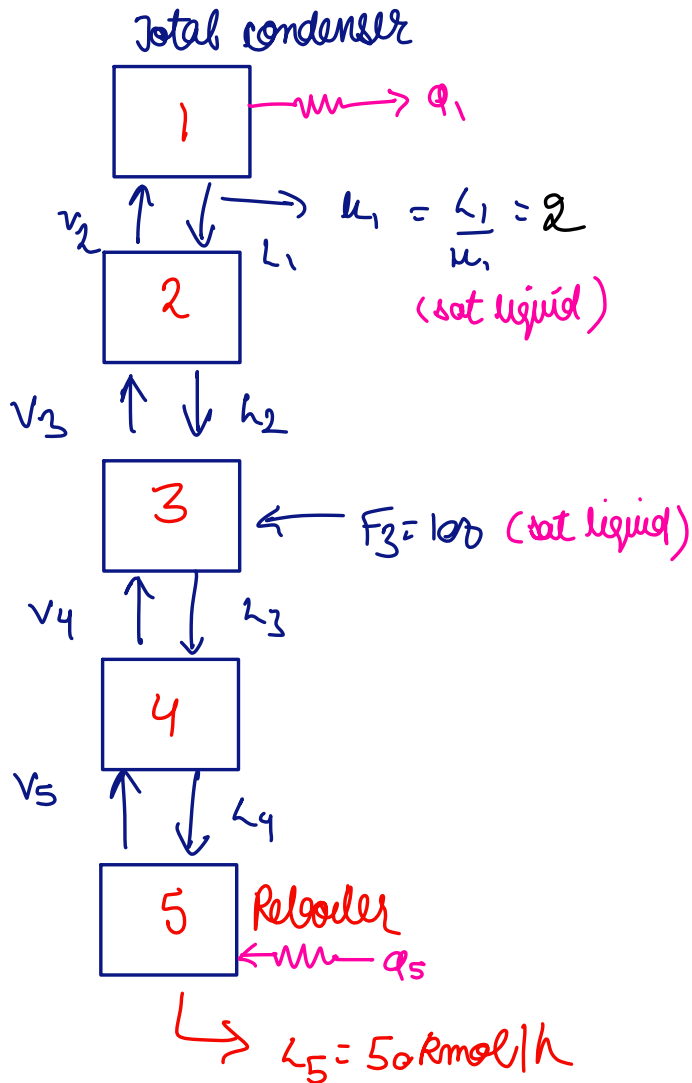
⑧ Compare the

both have to be satisfied individually $\leq \epsilon$
 $\epsilon \sim 10^{-2}$ (generally)

$$\sum_{i=1}^N \left[\left(\frac{T_j^R - T_j^{R+1}}{T_j^R} \right)^2 + \left(\frac{V_j^R - V_j^{R+1}}{V_j^R} \right)^2 \right] \leq \text{Tolerance}$$

summed over all trays

ex



comp $Z_{i,3}$

C_3 0.3

C_4 0.3

C_5 0.4

saturated liquid @ 7 bar

All stages operate at
 $P_f = 7 \text{ bar}$

Find tray compositions x_{ij} , y_{ij} & T_{ij}

From overall mass balance of column

$$F_3 = L_5 + V_1$$

$$V_1 = F_3 - L_5 = 50 \text{ kmol/h}$$

$$L_1 = 2V_1 = 100 \text{ kmol/h}$$

$$V_2 = L_1 + V_1 = 350 \text{ kmol/h}$$

$$V_2 = V_1 + V_1 = 100 + 50 = 150 \text{ kmol/h}$$

guess value of these variables T_j & V_j

so fixed, not guessing guess T_j & V_j (initial guess across different trays)

← obtained from mass balance	j	V_j	T_j	
	1	0	18	→ propane bp @ 7 bar [$j=1$, mostly propane logic or uska BP lelia]
	2	150	32	} linear interpolation
	3	150	46	
	4	150	60	
	5	150	74	→ bp n_C @ 7 bar [similar logic]

calculate L_j from mass balance

$$V_{j+1} + L_{j-1} = V_j + L_j$$

$$L_j = V_{j+1} + L_{j-1} - V_j$$

$$L_2 = 150 + L_1 - V_2$$

✓ In general

$$L_j = V_{j+1} - V_1 - \sum_{m=1}^j (F_m - V_m - W_m) \quad \left(\text{from overall mass balance @ } (1-j)^{\text{th}} \text{ tray} \right)$$

$$L_2 = V_3 - V_1 = 150 - 50 = 100$$

$$U_1 = F_3 - L_5 = 50$$

$$L_1 = 2U_1 = 100$$

$$V_1 = L_1 + U_1 = 150$$

$$L_3 = V_4 + F_3 + U_1 = 200$$

$$L_4 = V_5 + F - U_1$$

$$= 150 - 50 + 100 = 200 \text{ kmol/hr}$$

In this case, $V_1 = 0$ & $W_j = 0$

$$L_j = L_{j-1} = V_j + \sum_{m=1}^{j-1} (F_m - V_m)$$

$$L_5 = L_4 = V_5 + F_3 - U_1 = 150 + 100 - 50 = 200$$

$$L_4 = L_3 = V_4 + F_3 - U_1 = 150 + 100 - 50 = 200$$

$$L_3 = L_2 = V_3 - U_1 = 150 - 50 = 100$$

$$L_2 = L_1 = 100$$

construct TOM

$$A_j = L_{j-1}$$

$$B_{ij} = - \left[\left(V_j + \overset{0}{U_j} \right) K_{ij} + L_j + U_j \right]$$

$$C_{ij} = V_{j+1} K_{i,j+1}$$

$$D_{ij} = -Z_{ij} F_j$$

K_{ij} @ T_j & 7 var $\rightarrow$ estimate k_{ij} for T_j

since $V_1 = 0$ & $w_j = 0$

$$B_j = - \left[V_{j+1} + \underbrace{\sum_{m=1}^j (F_m - u_m)}_{L_j} + U_j + V_j K_{ij} \right]$$

stage j $\rightarrow$		18	32	46	60	74
species i		1	2	3	4	5
c_3	1	1.23	1.63	2.17	2.7	3.3
c_4	2	0.33	0.5	0.7	0.95	1.25
c_5	3	0.104	0.16	0.25	0.36	0.5

K increase, $T \uparrow$
 $K \downarrow$, $MW \uparrow$ } imp

$i=1$

$$C_j = V_{j+1} K_{1,j+1}$$

$$A_5 = L_4 = 200$$

$$A_4 = L_3 = 200$$

$$A_3 = L_2 = 100$$

$$A_2 = L_1 = 100$$

$$B_{1,5} = - [150 \times 3.3 + 50 + 50] = -545$$

$$B_{1,4} = - [150 \times 2.7 + 200] = -605$$

$$B_{1,3} = - [150 \times 2.17 + 200] = -525.25$$

$$B_{1,2} = - [150 \times 1.63 + 100] = -344.5$$

$$B_{1,1} = - [0 \times 1.23 + 100 + 50] = -150$$

$$C_{1,4} = 150 \times 3.3 = 495$$

$$C_{1,3} = 150 \times 2.7 = 405$$

$$C_{1,2} = 150 \times 2.17 = 325.5$$

$$C_{1,1} = 150 \times 1.63 = 244.5$$

Finally $D_j = F_j Z_{1,j} = D_3 = -100(0.3) = -30$

$$D_1 = D_2 = D_4 = D_5 = 0$$

TDM for $i=1$ (C_3)

$$\begin{bmatrix} B_{11} & C_{11} & 0 & 0 & 0 \\ A_2 & B_{12} & C_{12} & 0 & 0 \\ 0 & A_3 & B_{13} & C_{13} & 0 \\ 0 & 0 & A_4 & B_{14} & C_{14} \\ 0 & 0 & 0 & A_5 & B_{15} \end{bmatrix} \begin{bmatrix} x_{11} \\ x_{12} \\ x_{13} \\ x_{14} \\ x_{15} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ D_{13} \rightarrow -Z_{13}F_3 \\ 0 \\ 0 \end{bmatrix}$$

$$p_i = \frac{D_i - A_i p_{i-1}}{B_i - A_i q_{i-1}}$$

$$q_i = \frac{C_i}{B_i - A_i q_{i-1}}$$

$$p_1 = \frac{D_{10}}{B_{11}} = 0$$

$$p_2 = \frac{\overset{0}{D_2} - \overset{100}{A_2} \overset{0}{p_1}}{\underset{-30}{B_2} - \underset{100}{A_2} \underset{0}{q_1}} = 0$$

$$p_3 = \frac{\overset{-30}{D_3} - \overset{100}{A_3} \overset{0}{p_2}}{\underset{-525.25}{B_3} - \underset{100}{A_3} \times \underset{-0.945}{q_2}} = 0.087$$

$$p_4 = \frac{\overset{0}{D_4} - \overset{200}{A_4} \overset{0.087}{p_3}}{\underset{-545}{B_4} - \underset{200}{A_4} \underset{-1.33}{q_3}} = 0.047$$

$$p_5 = \frac{\overset{0}{D_5} - \overset{200}{A_5} \overset{0.047}{p_4}}{\underset{-545}{B_5} - \underset{200}{A_5} \underset{-1.33}{q_4}} = 0.034$$

general expression for Thomas algorithm

$$p_j = \frac{C_j}{B_j - A_j p_{j-1}}$$

$$q_j = \frac{D_j - A_j q_{j-1}}{B_j - A_j p_{j-1}}$$

$$x_j = q_j - p_j x_{j+1}$$

$$q_1 = \frac{C_{11}}{B_{11}} = \frac{244.5}{-150}$$

$$q_2 = \frac{C_{12}}{B_{12} + 100 \times 1.63} = \frac{325.5}{-344.5 + 163} = -1.79$$

$$q_3 = \frac{C_{13}}{B_{13} - A_3 q_2} = -1.17$$

$$q_4 = \frac{C_{14}}{B_{14} - A_4 q_3} = -1.33$$

$$x_5 = p_5 = 0.034$$

$$x_4 = p_4 - q_4 x_5 = 0.092$$

$$x_3 = p_3 - q_3 x_4 = 0.1953$$

$$x_2 = p_2 - q_2 x_3 = 0.35$$

$$x_1 = p_1 - q_1 x_2 = 0.57$$

stage $j \rightarrow$	$x \rightarrow$				
	1	2	3	4	5
c_3 ($i=1$)	0.57	0.35	0.195	0.092	0.034
c_4 ($i=2$)	0.19	0.38	0.45	0.49	0.41
c_5 ($i=3$)	0.018	0.115	0.33	0.48	0.78
$\sum_{i=1}^3 x_{ij}$	0.778	0.845	0.775	1.062	1.225

$$\bar{x}_{ij} = \frac{x_{ij}}{\sum_{i=1}^3 x_{ij}}$$

$$\sum_{i=1}^N y_{ij} = 1 \Rightarrow \sum_{i=1}^3 k_{ij} \bar{x}_{ij} = 1 \rightarrow \text{for each tray}$$

$$k_{1j} \bar{x}_{1j} + k_{2j} \bar{x}_{2j} + k_{3j} \bar{x}_{3j} = 1 \quad T_j^{k+1} = b_1(T_j) \bar{x}_{1,j} + b_2(T_j) \bar{x}_{2,j} + b_3(T_j) \bar{x}_{3,j} = 1$$

$\rightarrow$ Calculate T_j^{k+1}

Scaled Values (x_{ij})

	$j=1$	2	3	4	5	Calculate
$i=1$	0.733	0.41				$\left \frac{T_j^{k+1} - T_j^k}{T_j^k} \right < 0.01$
$i=2$	0.244					
$i=3$	0.023					

Once you know T_j , use enthalpy balance to calculate V_j

$$\text{Then check } \frac{V_j^{k+1} - V_j^k}{V_j^{k+1}} < 0.01$$

T	K_i
✓	✓
✓	✓
✓	✓
✓	✓
✓	✓
✓	✓

$$K_i = \alpha_i + B_i T_j$$

$$\sum \left[\frac{T_j^R - T_j^{R+1}}{T_j^R} \right]^2 \leq \text{tolerance}$$

$V_j \equiv$ Enthalpy eqn

$$h_v(T)$$

$$h_l(T)$$

$$h_j$$

$$V_j$$

$$\sum \left[\frac{V_j^R - V_j^{R+1}}{V_j^R} \right]^2 < \text{Tolerance}$$

$$\sum_{j=1}^N \left[\frac{T_j^R - T_j^{R+1}}{T_j^R} \right]^2 + \sum_{j=1}^N \left[\frac{V_j^R - V_j^{R+1}}{V_j^R} \right]^2 \leq \text{Tolerance}$$

Steps

1) Identify the tear variables & assume its value

2) Calculate K_{ij} values

3) Calculate h_j values & (A, B, C of the TDM) $\rightarrow$ specific for a species

4) Solve the TDM for each species i

5) Normalise α_{ij}

6) Estimate T_j from $\sum_{i=1}^C y_{ij} = 1$

7) Evaluate V_j & check for convergence

$$K_i = \frac{P^S}{P} \rightarrow \text{Clausius Clapeyron eqn}$$

8/2/24

Multicomponent multistage distillation

(Bubble point method)

(Tear variable)

T_j, V_j

$\hookrightarrow K$

A, B, C

TDM

x_{ij}

$x_{ij} \rightarrow$ calculated TDM solution

$$\sum_{i=1}^C x_{ij} \neq 1$$

$\rightarrow$ Normalising to ensure sum is equal to 1

$$\bar{x}_{ij} = \frac{x_{ij}}{\sum_{i=1}^C x_{ij}} \Rightarrow \sum \bar{x}_{ij} = 1$$

$$\sum y_{ij} = 1 = \sum_{i=1}^C K_{ij} \bar{x}_{ij}$$

$\hookrightarrow f_i(T_j)$

$$y = Kx$$

P^s can be estimated from
Clausius Clapeyron
Equation

$$R_i = P^s / P$$

$\hookrightarrow$ operating pressure

$$\begin{array}{c|c} T & K_i \\ \hline \checkmark & \checkmark \\ \checkmark & \checkmark \\ \checkmark & \checkmark \end{array}$$

$$K_i = \alpha_i + B_i T$$

$$\sum_{j=1}^N \left[\frac{T_j^k - T_j^{k+1}}{T_j^{k+1}} \right]^2 \leq \text{Tolerance}$$

$V_j =$ enthalpy equation

$$h_v(T) \quad L_j$$

$$h_e(T) \quad U_j$$

$$\sum_{j=1}^N \left[\frac{V_j^k - V_j^{k+1}}{V_j^k} \right]^2 < \text{Tolerance}$$

$$\sum_{j=1}^N \left(\frac{V_j^R - V_j^{R+1}}{V_j^R} \right)^2 + \sum_{j=1}^N \left(\frac{T_j^R - T_j^{R+1}}{T_j^R} \right)^2 \leq \text{Tolerance}$$

Steps

1. Identify the trace variables & assume its values
2. Calculate K_{ij} values
3. Calculate L_j values, & (A, B, C of the TDM) $\rightarrow$ specific for a species
4. Solve the TDM for each species i
5. Normalise x_{ij}
6. Estimate τ_j from $\sum_{i=1}^C y_{ij} = 1$
7. Evaluate v_{ij} & check for convergence

Simultaneous Correction Method

Material balance (M)

Input $\equiv$

$$(y_{ij+1} v_{j+1} + x_{ij-1} l_{j-1} + z_{ij} f_j)$$

Output $\equiv$

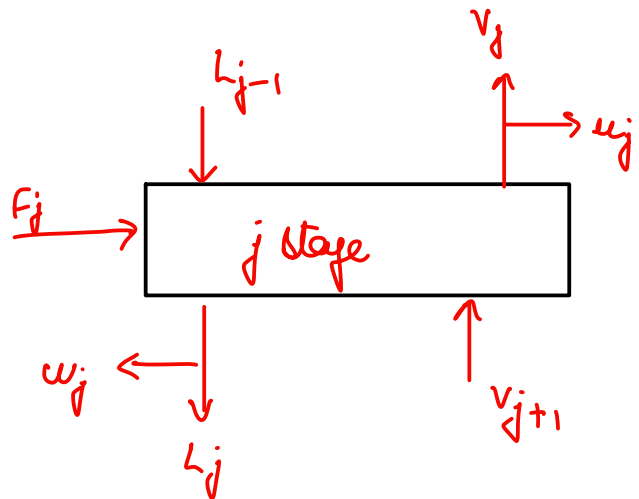
$$(u_j + v_j) y_{ij} + (w_j + l_j) x_{ij}$$

$$M \equiv \text{Input} - \text{Output} = 0$$

$$\Rightarrow \underbrace{\left(1 + \frac{u_j}{v_j}\right)}_{\gamma_{vj}} \underbrace{\left(\frac{v_j}{v_j}\right)}_{v_{ij}} y_{ij} + \underbrace{\left(1 + \frac{w_j}{l_j}\right)}_{\gamma_{lj}} \underbrace{\left(\frac{l_j}{l_j}\right)}_{l_{ij}} x_{ij} - \underbrace{y_{ij+1} v_{j+1}}_{v_{ij+1}} - \underbrace{x_{ij-1} l_{j-1}}_{l_{ij-1}} - \underbrace{z_{ij} f_j}_{b_{ij}} = 0$$

$\rightarrow$ Mij equation

$$(1 + \gamma_{vj}) v_{ij} + (1 + \gamma_{lj}) l_{ij} - v_{ij+1} - l_{ij-1} - b_{ij} = 0$$



$$v_{ij} = y_{ij} v_j$$

$$l_{ij} = x_{ij} h_j$$

$$v_j = \sum_{i=1}^C v_{ij}$$

$$h_j = \sum_{i=1}^C l_{ij}$$

So,

$$y_{ij} = k_{ij} x_{ij}$$

$$\frac{v_{ij}}{\sum_{i=1}^C v_{ij}} = k_{ij} \frac{l_{ij}}{\sum_{i=1}^C l_{ij}}$$

→ Eij equation

$$k_{ij} l_{ij} \frac{\sum_{i=1}^C v_{ij}}{\sum_{i=1}^C l_{ij}} - v_{ij} = 0$$

$h_v, h_L \rightarrow$ function of temperatures
so not unknown

Enthalpy equation

$$(1+y_{vj}) h_{vj} v_j + (1+y_{Lj}) h_{Lj} h_j - h_{vj+1} v_{j+1} - h_{Lj-1} h_{j-1} - h_{Fj} F_j = 0$$

Hij equation

$$(1+y_{vj}) h_{vj} \sum_{i=1}^C v_{ij} + (1+y_{Lj}) h_{Lj} \sum_{i=1}^C l_{ij} - h_{vj+1} \sum_{i=1}^C v_{ij+1} - h_{Lj-1} \sum_{i=1}^C l_{ij-1} - h_{Fj} \sum_{i=1}^C f_{ij} = 0$$

C mass balance equations

C $y = Kx$ "

1 enthalpy equation

eqns = $2C+1$ per stage

$\pm Q_j$

↳ external heat
for every stage

y_v, y_L, Φ, P

we are setting (degrees of freedom)

unknowns v_{ij}, l_{ij}, T_j
(C) (C) (1)

$2C+1$

→ if anything other than this unknown
then system won't be closed

→ vector of unknowns

$$\bar{x}_j = [v_{1j}, v_{2j}, v_{3j}, \dots, v_{cj}, T_j, l_{1j}, l_{2j}, \dots, l_{cj}]^T \quad (2c+1) \text{ unknown variables}$$

$$\bar{F}_j = [H_j, M_{1j}, M_{2j}, \dots, M_{cj}, E_{1j}, E_{2j}, \dots, E_{cj}]^T \quad (2c+1) \text{ equations}$$

→ vector of equations

Roots of $\bar{F}(\bar{x}) = 0$

→ Newton Raphson

$$\bar{x}^{m+1} = \bar{x}^m - \alpha F(\bar{x}) \left(\frac{d\bar{F}}{d\bar{x}} \right)^{-1}$$

→ we can't solve any stage explicitly as it will be linked to previous stage, so we need to solve together

controlling factor, to prevent divergence, as system is highly non linear

$$\frac{d\bar{F}}{d\bar{x}} = \begin{bmatrix} \frac{dF_1}{dx_1} & \frac{dF_1}{dx_2} & \dots & \frac{dF_1}{dx_n} \\ \frac{dF_2}{dx_1} & \frac{dF_2}{dx_2} & \dots & \frac{dF_2}{dx_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{dF_N}{dx_1} & - & - & \frac{dF_N}{dx_n} \end{bmatrix} \quad \begin{matrix} \text{Block TDM} \\ N \times N \end{matrix}$$

where $\frac{dF_j}{dx_j}$ is a matrix of $(2c+1) \times (2c+1)$

Except $\underbrace{\frac{dF_j}{dx_{j-1}}}_A, \underbrace{\frac{dF_j}{dx_j}}_B, \underbrace{\frac{dF_j}{dx_{j+1}}}_C$ all other are zero
→ Blocks A, B, C

$$A_j = \frac{dF_j}{dx_{j-1}}$$

$$v_{1,j-1} \dots v_{c,j-1}, T_{j-1}, l_{1,j-1}, l_{2,j-1} \dots l_{c,j-1}$$

$$\begin{bmatrix} H_j \\ M_j \\ \vdots \\ M_{c,j} \\ E_j \\ \vdots \\ E_{c,j} \end{bmatrix} \begin{bmatrix} + & \dots & \dots & \dots & \dots \\ -1 & \dots & \dots & \dots & \dots \\ & \dots & \dots & \dots & \dots \\ & & \dots & \dots & \dots \\ & & & -1 & \text{(till } M_{c,j}) \end{bmatrix}$$

$$B_j = \frac{dF_j}{dx_j}$$

$$v_{1,j} \dots v_{c,j}, T_j, l_{1,j} \dots l_{c,j}$$

$$\begin{bmatrix} H_j \\ M_j \\ \vdots \\ M_{c,j} \\ E_j \\ \vdots \\ E_{c,j} \end{bmatrix} \begin{bmatrix} + & \dots & \dots & \dots & \dots & \dots & + \\ + & \dots & \dots & \dots & + & \dots & \dots \\ & \dots & \dots & \dots & \dots & \dots & \dots \\ & & \dots & + & \dots & \dots & \dots \\ + & \dots & \dots & \dots & \dots & \dots & + \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ + & \dots & \dots & \dots & \dots & \dots & + \end{bmatrix}$$

$$c_j = \frac{dF_j}{dx_{j+1}}$$

$$v_{1j+1} \dots v_{cj+1}, \tau_{j+1}, \ell_{1j+1} \dots \ell_{cj+1}$$

$$\begin{bmatrix} N_{ij} \\ M_{ij} \\ \vdots \\ M_{cj} \\ E_{ij} \\ \vdots \\ E_{cj} \end{bmatrix} \begin{bmatrix} + & - & - & - & - & + \\ + & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & + \end{bmatrix} \begin{bmatrix} v_{1j+1} \dots v_{cj+1}, \tau_{j+1}, \ell_{1j+1} \dots \ell_{cj+1} \end{bmatrix}$$

KGF ki MKS, Tiwari BKL hai
can we be at peace???

Block Tridiagonal Matrix

$$\begin{bmatrix} \bar{B}_1 & \bar{C}_1 & 0 & \dots & 0 & 0 \\ \bar{A}_2 & \bar{B}_2 & \bar{C}_2 & \dots & 0 & 0 \\ & & & \dots & & \\ & & & & \bar{A}_m & \bar{B}_m \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \vdots \\ \vdots \\ \vdots \\ \Delta x_m \end{bmatrix} = \alpha \begin{bmatrix} -\bar{r}_1 \\ -\bar{r}_2 \\ \vdots \\ \vdots \\ \vdots \\ -\bar{r}_m \end{bmatrix}$$

$\left(\frac{dF}{dX} \right)$
 (ΔX)
 (F)

$$\frac{\Delta X}{\alpha} = F \left(\frac{dF}{dX} \right)^{-1}$$

$$\left(\frac{dF}{dX} \right) (\Delta X) = -F\alpha$$

here while solving BTOM
we also solve individual
matrices, so one shot solution
(simultaneously solving)

Stage 1 :
(First Row)

(First Row) $B_1 \leftarrow I$

(First Row) $B_1 \leftarrow I$
 $c \leftarrow B_1^{-1} c_1$

$$F_1 \leftarrow -B_1^{-1} F_1$$

New elements after implementing steps similar to Thomas algorithm

$$\begin{array}{ccccccccc} b_1 & c_1 & & - & - & - & & 0 & x_1 & d_1 \\ a_2 & b_2 & c_2 & & - & - & - & 0 & x_2 & d_2 \\ , & & & & & & & & , & , \\ , & & & & & & & & , & , \\ , & & & & & & & & , & , \\ , & & & & & & & & , & , \\ , & & & & & & & & , & , \\ & - & - & - & a_m & b_m & & & x_m & d_m \end{array}$$

$$b_1 x_1 + c_1 x_2 = d_1$$

$$x_1 + \frac{c_1}{b_1} x_2 = \frac{d_1}{b_1}$$

$$a_2 x_1 + b_2 x_2 + c_2 x_3 = d_2$$

Stage 2: Tell $N-1$ row

$$\bar{c}_j \leftarrow (\bar{b}_j - \bar{a}_j c_{j-1})^{-1} \bar{c}_j \quad \bar{a}_j \leftarrow a$$

$$F_j \leftarrow -(B_j - A_j C_{j-1})^{-1} (F_j - A_j F_{j-1})$$

$$B_j \leftarrow I$$

last Row, N

$$B_N \leftarrow I$$

$$A_N \leftarrow 0 \quad (C_N \text{ vanish})$$

$$F_N \leftarrow -(B_N - A_N C_{N-1})^{-1} (F_N - A_N F_{N-1})$$

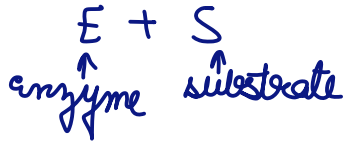
$$c_N \leftarrow 0$$

29/2

→ biocatalyst

Enzyme → takes part in rxn

Catalyst → doesn't " "



→ Enzyme do bind with substrate, to form an intermediate then product

Enzyme rxns are never single reaction

↳ always intermediate formed



Substrate → Product



In enzymatic rxn

$S \rightarrow P$ x, will never happen

In catalytic rxn, it can happen, without catalyst

Absorption → no new formation, no transformation

Chemical rxn → formation of a new product

In enzymic reaction → one chemical rxn, followed by another rxn

" chemical " → first adsorption, then reaction



$$\frac{d[ES]}{dt} = k_1[E][S] - k_{-1}[ES] - k_2[ES] \quad \text{--- ①}$$

$$\frac{d[S]}{dt} = -k_1[S][E] + k_{-1}[ES] \quad \text{--- ②}$$

$$\frac{d[P]}{dt} = k_2[ES] \quad \text{--- ③}$$

→ 4 eqns, 4 unknowns (species)

3 ODE
1 equation

$[E]_t + [ES]_t = \text{const}$ → mole conservation → total enzyme at any time is conserved

ICs

$$[ES] = 0 \text{ @ } t=0 \quad [S] = S_0 \text{ @ } t=0 \quad [P] = 0 \text{ @ } t=0$$

$$[ES] = \frac{K_1[S][E]}{K_{-1} + K_2} \quad \text{--- (1)}$$

$\frac{d[ES]}{dt} = 0 \rightarrow$ at maximum, this is desired

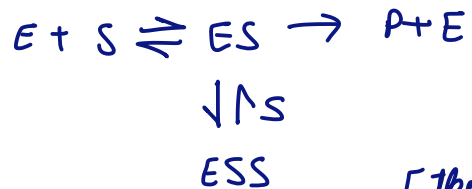
$$\frac{d[S]}{dt} = -K_1[S][E] + \frac{K_{-1}K_1[S][E]}{K_{-1} + K_2} \quad \text{--- (2)}$$

$$\frac{d[P]}{dt} = \frac{K_1K_2[S][E]}{K_{-1} + K_2} \quad \text{--- (3)}$$

$\rightarrow$ converted two step rxn into a single step rxn

Factors affecting $\rightarrow$ fever in body + digestion example

1. Temperature
2. If substrate is in excess
(similarly with if P is excess)



[this may happen]

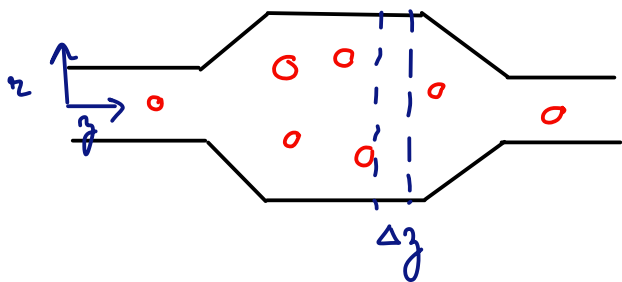
Inhibition $\rightarrow$ Neurotoxin

Competitive	Non-competitive	Uncompetitive
Substrate Inhibitor Compete for active site	Different site	
$E + S \rightleftharpoons ES \rightarrow E + P$ $+ I$ $\uparrow \downarrow$ EI	$E + S \rightleftharpoons ES \rightarrow E + P$ $+ I$ $\uparrow \downarrow$ $EI + S \rightarrow EI/S$	$E + S \rightleftharpoons ES \rightarrow E + P$ $+ I$ $\uparrow \downarrow$ EI/S

$\rightarrow$ Earlier shape for I was not there but after S attached, shape / morphology changed and I can also attach to enzyme molecule

11/3/24

Packed Bed Reactor



Reaction and diffusion / convection happen simultaneously
 how to model performance / design scale-up of the system?

compressible / Incompressible fluid : S depends on P
 " " flow : v vs speed of sound in that medium

Assumptions

1. Isothermal operation
2. Steady state
3. No fluctuation
4. No radial transport
5. Incompressible fluid & flow
6. Convection dominates diffusion

across Δz , shell balance → reaction rate per unit volume

$$N_z|_z - N_z|_{z+\Delta z} + r_A Q_c(\Delta z) = 0$$

→ specific surface area per unit volume of effective catalyst surface area ($m^2 \text{ s.a.} / m^3 \text{ bed vol}$)

$$N_z = \left(-D \frac{\partial C_A}{\partial z} + v C_A \right) A = v C_A A$$

$$v C_A|_z - v C_A|_{z+\Delta z} + r_A Q_c(\Delta z) A = 0 \Rightarrow -\frac{\partial}{\partial z} (v C_A) + r_A Q_c = 0$$

How does one describe flow in porous media?

→ Darcy's law, Ergun Eqn

$$v = \frac{-k}{\mu} \Delta p = \frac{-k}{\mu} \frac{\partial p}{\partial z}$$

$$v \neq f(z) \therefore \frac{\partial p}{\partial z} = \text{constant}$$

effect of fully developed flow

$$v \frac{\partial C_A}{\partial z} = r_A a_c$$

Formation of MTBL because A is being depleted at catalyst surface

Microscale as well as macroscale steady state

Rate of reaction = Rate of MT
 $\therefore$ no Damköhler term
 $-r_A = k(C_A - C_{A_s})$

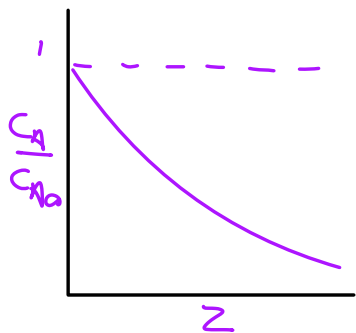
But $C_A \gg C_{A_s}$ so $-r_A = k C_A$

$\hookrightarrow$ MTC, not reaction rate constant

Let $C_{A0} = C_A(z=0)$

so, $v \frac{\partial C_A}{\partial z} = -k C_A a_c \Rightarrow \int_{C_{A0}}^{C_A} \frac{\partial C_A}{C_A} = -\frac{k a_c}{v} \int_0^z dz \Rightarrow \ln \frac{C_A}{C_{A0}} = -\frac{k a_c z}{v}$

$$C_A = C_{A0} \exp\left(-\frac{k a_c z}{v}\right)$$



Estimation of MTC: Sherwood Number correlation
 $Sh \sim 2 + f(Re, Sc) \sim 2 \text{ to } 3$

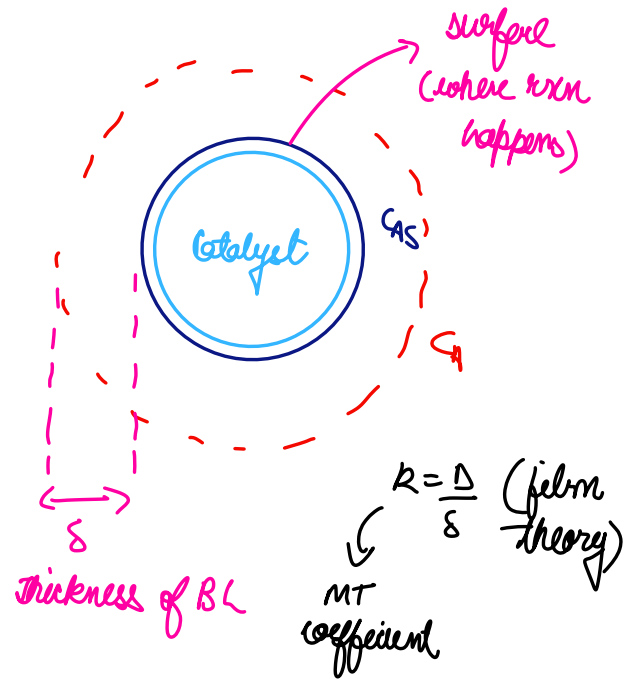
If diffusion term isn't neglected

$$N_z = \left(-D \frac{\partial C_A}{\partial z} + v C_A \right) A$$

Then $D \frac{\partial^2 C_A}{\partial z^2} - v \frac{\partial C_A}{\partial z} - k C_A a_c = 0$

Non dimensionalization: $z^* = z/L$ $C_A^* = C_A/C_{A0}$

Then $\frac{D C_{A0}}{L^2} \frac{\partial^2 C_A^*}{\partial z^{*2}} - \frac{v C_{A0}}{L} \frac{\partial C_A^*}{\partial z^*} - k C_{A0} C_A^* a_c = 0$



$$\Rightarrow \frac{D}{Vh} C_A^* \frac{d^2 z^+}{dz^{+2}} - C_A^* \frac{d^2 z^+}{dz^{+2}} - \frac{k_{ac} L}{V} C_A^* = 0$$

$$\frac{1}{Pe} C_A^* \frac{d^2 z^+}{dz^{+2}} - C_A^* \frac{d^2 z^+}{dz^{+2}} - D_{A1} C_A^* = 0$$

↳ Damkohler No. of 1st kind

Boundary conditions

$$\left. \begin{aligned} C_A^* (z^+ = 0) &= 1 \\ \frac{\delta C_A^*}{\delta z^+} (z^+ = 1) &= 0 \end{aligned} \right\} \text{open BC showing that no further reaction happens during exit}$$

$$Pe = \frac{\text{convection}}{\text{diffusion}} \quad \text{so when } L \gg d \quad \frac{1}{Pe} \rightarrow 0 \Rightarrow \text{d term vanished} \quad (Pe \gg 1)$$

$$Da_2 = Pe \times Da_1 \Rightarrow \frac{k_{ac} L}{V} \times \frac{D}{V L} = \frac{k_{ac} D}{V^2}$$

If $Pe \sim \frac{1}{Da}$ and both $\ll 1$, Reaction + Diffusion system (no convection)

If $Da \ll 1$ no reaction, diffusion + convection, system is MT-limited
 Reaction
 MT

If $Da \gg 1 \rightarrow$ Reaction is dominant mode of Transport, little MT happens
 (Packed bed size / pores etc. doesn't matter)

$$\frac{\partial C}{\partial t} + u \nabla C = D \nabla^2 C$$

& (dispersion coefficient) $D_{\text{porous}} + u_0 h C$

$$\frac{\partial C}{\partial t} + u \nabla C = \gamma \nabla^2 C$$

$$\frac{\partial C}{\partial t} = \gamma \nabla^2 C$$

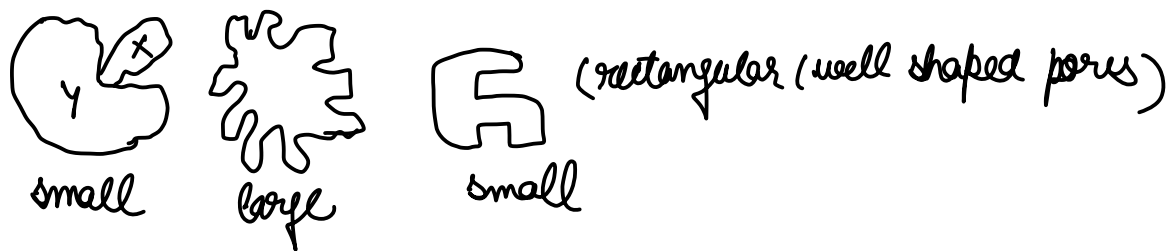
$$D_{\text{porous}} \sim \phi D_{\text{bulk}}$$

(considered diffusively)

In pebble number $D = D_{\text{porous}}$

"Showerood" $D = D_{\text{bulk}}$

catalyst particles are porous $\phi_p \rightarrow$ can be large / small



If ϕ_p is large $\rightarrow$ homogeneous diffusion / reaction inside particle

$r-\theta-\phi$ (spherical coordinate system for pore)

$$4\pi r^2 (N_{r+\Delta r} - N_r) + r_A 4\pi r^2 \Delta r = 0$$

$$N_r = -D \frac{\partial C_A}{\partial r}$$

$$\text{so, } \frac{D}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial C_A}{\partial r} \right) + r_A = 0$$

$$D_p \sim \phi D_b$$

$$\downarrow$$

$$-k_{\text{rxn}} C_A \text{ (1st order assumption)}$$

$$\text{at } r=0 \quad \frac{\partial C_A}{\partial r} = 0 \quad \therefore \text{radial symmetry}$$

$$\text{at } r=R \quad C_{A,r} = C_{A,s}$$

$$\text{at } r=R_p: \quad -D \frac{\partial C_{A,r}}{\partial r} = K_{MT} (C_A - C_{A,r})_{r=R_p}$$